

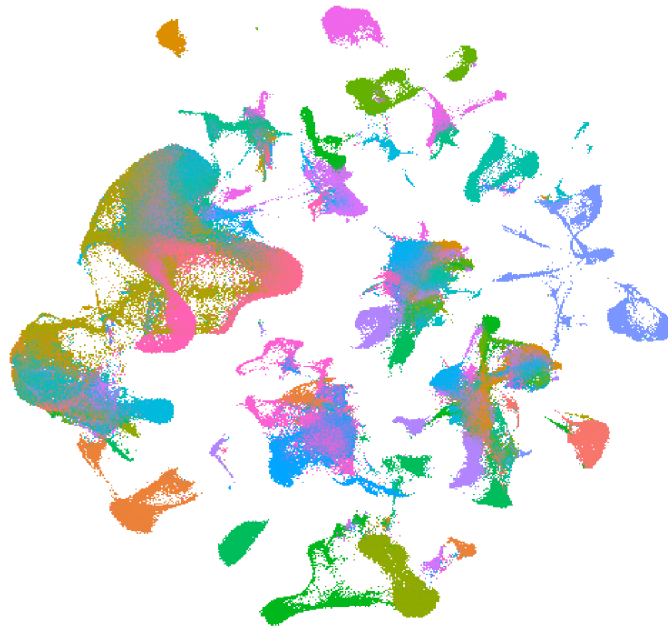


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Cell Type Annotation in Transcriptionally Homogeneous Single-Cell Data

*An Evaltion of Differential Expression Analysis and Advanced Embedding
Methods*



Master Thesis

By

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Declaration on use of generative AI in this report

We used it for idea generation and design of the experimental set up. It was also used to help explain and clarify difficult concepts. Finally, it was also used to help assist coding in both Python and R. We used the internal AI provided by Novo Nordisk: NNGPT.

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Abstract

Single-cell RNA sequencing (scRNA-seq) has become an integrated part of quality control in stem cell research, but standard differential expression (DE) analysis may fail to distinguish closely related cell types in transcriptionally homogeneous samples. This thesis addresses two questions: (1) How does transcriptional homogeneity affect DE-based cell type identification? (2) Can alternative embedding methods improve discrimination in homogeneous settings? In Part 1 of this thesis, we examine the effect of increasing transcriptional homogeneity on differential expression analysis. By subsetting a large cell atlas into smaller groups of varying homogeneity based on average Pearson correlation between cell types, we confirmed that DE analysis performance degrades substantially in homogeneous samples. These findings motivated Part 2, where we investigate whether alternative embedding methods can improve cell type discrimination in such challenging settings. In Part 2, we compared four data representations: highly variable genes (HVG) from log-normalized counts, principal component analysis (PCA) embeddings, variational autoencoder (VAE) latent embeddings, and VAE embeddings refined by latent diffusion model (LDM) denoising. Performance was evaluated using cell-type Local Inverse Simpson Index (cLISI) scores and label transfer accuracy via multinomial logistic regression. Contrary to expectations, PCA consistently performed similar or better than the more complex methods, with LDM providing minimal to no improvement over VAE, across both homogeneous and heterogeneous settings. These results suggest that improved representation learning alone is insufficient to overcome limitations in cell type discrimination, highlighting the role of annotation practices, evaluation frameworks and the lack of biological context in current modeling framework.

1 Introduction

Stem cells play an important role in the development of cell-based therapies, where differentiation protocols are used to generate specific target cell types. A central challenge in this setting is quality control: determining whether the resulting cell population corresponds to the intended cell type with sufficient purity. Reliable cell identity verification is therefore a critical component of both experimental and industrial stem cell applications, with single-cell sequencing technologies playing an increasingly prominent role in this process [1].

1.1 Single-cell RNA sequencing for cell identity characterization

In cells, RNA expression reflects the active transcriptional program and therefore serves as a proxy for cell identity and function [2]. Single-cell RNA sequencing (scRNA-seq) is a key technology for characterizing cell identities at high resolution. By measuring gene expression at the level of individual cells, scRNA-seq enables the profiling of thousands of cells across diverse tissues, developmental stages, or experimental conditions. Figure 1 illustrates the generation of scRNA-seq data, where transcript counts from individual cells are aggregated into an expression matrix of gene and cells.

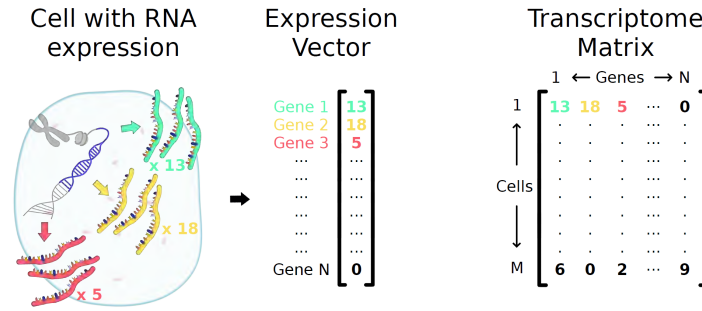


Figure 1: Schematic overview of scRNA data generation. RNA transcripts are counted in individual cells, and the combined transcript counts across many cells form the single-cell RNA (scRNA) data structure used throughout this thesis. source: [3]

In typical analysis workflows, cells are clustered by Luvain or Leiden clustering [4, 5] based on similarity in their expression profiles, with each cluster interpreted as a distinct cell type or cellular state. The biological interpretation of these groups typically relies on differential gene expression (DE) analysis, which is used to identify marker genes that distinguish one cluster from others. Marker genes can then be used to identify the cells. Traditionally, the annotation of cells from marker genes has been done by domain experts. More recently, computational approaches such as Gene Set Enrichment Analysis (GSEA) have been adopted. In GSEA, identified marker genes of a group of cells are compared to a curated database of known sets of marker gene representing established cell types and cellular states.

1.2 Context dependence of differential expression analysis

Recent work, however, has highlighted that DE analysis is **context-dependent**: the set of genes identified as markers depends strongly on the reference populations used for comparison [6]. As a result, the same cell type may appear to exhibit different marker signatures depending on which other cell types are included in the analysis.

This dependency on reference context has important consequences for cell type annotation. Comparing a cell population against closely related cell types may reveal generic transcriptional differences, while comparisons against unrelated populations tend to emphasize more cell-type-specific features. Consequently, the choice of comparison not only affects the resolution of the resulting classification, but also determines the biological interpretability of the identified markers.

Methods have been developed to address the lack of suitable background reference cells in query datasets for cell type classification. One approach identifies marker genes via differential expression (DE) analysis while mitigating the absence of an appropriate reference by combining scRNA-seq

datasets through *data integration*. However, integration typically restricts analyses to features shared across datasets, discarding dataset-specific features and thereby losing valuable information. This can bias results toward major, commonly shared cell types rather than rare or dataset-specific populations [7]. A second widely used approach is label transfer, in which a classifier trained on a reference dataset is applied to annotate unlabeled query cells. The key limitation is that labels are constrained to those present in the reference: novel or underrepresented cell states in the query cannot be discovered and may be misclassified.

These challenges are particularly pronounced in industrial stem cell differentiation pipelines, such as those aimed at producing pancreatic endocrine cells for the treatment of type 1 diabetes. In this setting, target cell types such as *beta cells*, may be present in the final sample across multiple developmental states and often share highly similar transcriptional profiles. When DE analysis is applied within such homogeneous populations, genuine biological differences may be too subtle to detect. Instead, the analysis may yield weak or non-specific markers, including broadly expressed housekeeping genes. This makes it difficult to robustly assess whether the desired cell type has been successfully generated, motivating a systematic investigation of DE performance under varying levels of transcriptional homogeneity.

1.3 Why embeddings are central to single-cell analysis

Raw scRNA-seq data present significant computational challenges: they are high-dimensional (typically 20,000+ genes), sparse with many zeros, and affected by technical noise. Direct analysis of such high-dimensional data suffers from the *curse of dimensionality*, where distances between points become increasingly similar, reducing contrast between nearest and farthest neighbors and thereby degrading clustering performance. [8].

Low-dimensional latent representations or *embeddings* address these challenges by providing a more computationally tractable and continuous space for learning biological structure [9]. However, the utility of these embeddings extends beyond computational efficiency, as they directly determine how cells are grouped for downstream analyses. Although differential expression (DE) analysis is performed on normalized gene expression counts, its outcome depends critically on how cells are clustered for comparison—and these groupings are almost always derived from embeddings such as PCA or other latent representations [4, 9].

The quality of the embedding therefore decisively influences DE outcomes. A *good* embedding preserves neighborhoods from the original space, meaning cells close together in the embedding should have similar expression profiles [10]. If neighborhood preservation fails, biologically distinct populations may be poorly separated or mixed during clustering, causing differential expression signals to be diluted or lost. Consequently, refining an embedding—rather than modifying gene expression values directly—improves the organization of cells in latent space, enabling more coherent clustering and more informative differential expression contrasts.

This dependency on good embeddings is especially relevant in a transcriptionally homogeneous setting, where biological differences between cell types are subtle and the variance is often dominated by noise rather than biological signal. A linear method like PCA might fail to separate closely related cell types or developmental states in these conditions, which could lead to mixed or poor clusters and therefore weak or misleading DE results [8]. These limitations are the motivation for exploring other alternative embedding strategies that better capture subtle biological structure in homogeneous conditions.

An effective embedding strategy should ideally be able to reduce technical noise while preserving biological meaningful differences between cells, such that grouping of cells in this resulting representation would give biologically meaningful groupings. A range of methods to improve representations of scRNA-seq data exists, broadly aiming to stabilize variance, mitigate technical effects, or learn latent structure from noisy measurements like *Harmony*, *scTransform* or *scVI* [11, 12, 13]. While such methods have been shown to improve global organization and batch mixing in many settings, their behavior in highly homogeneous populations, where biological variation is subtle, is less well characterized.

1.4 Diffusion based models in scRNA

In the realm of deep generative modeling, diffusion denoising probabilistic models (DDPM) based models have been shown as powerful tools for denoising. DDPMs iteratively remove noise from data by learning to reverse a diffusion process in which noise (often gaussian) is iteratively added. DDPMs has proven strong performance within fields like text, sound and especially image generation, where diffusion models have shown great success in denoising low quality images to high quality images [14, 15, 16]. These successes suggest that diffusion models might be capable of refining learned representations by projecting noisy inputs toward a structured manifold. However, it is not obvious that these properties transfer to transcriptomics data, which differ substantially from the other fields in terms of noise structure and their generative process.

Several diffusion-based models have recently been proposed for scRNA-seq data, including scDiffusion, scLDM, and scVAED [17, 18, 19]. To date, these models have primarily been developed for data generation and simulation rather than for improving representations used in downstream analyses. In particular, the potential of diffusion models to refine latent embeddings in a way that improves cell grouping and subsequent DE analysis has not been systematically evaluated.

All the diffusion models referenced above are latent diffusion models (LDM). It is a type of model where a DDPM-style denoising process is applied to a latent representation of the data rather than directly on the raw gene expression data [20]. The architectural design choice of LDM is motivated by several properties of scRNA-seq data. First of all, the feature space of scRNA-seq data is extremely high dimensional and DDPM scale poorly for high dimensions. Therefore, dimensional reductions provide a more computational tractable domain where a diffusion process can be learned effectively. Secondly, scRNA-seq data is non Gaussian and DDPM models assume Gaussian noise. Latent representations learned by models such as variational autoencoders (VAE) are continuous and approximately Gaussian making them more compatible with a DDPM. The latent representations the DDPM is fitted to can vary. In the examples provided in the latter section, scDiffusion utilizes a foundation model [17], scLDM uses a convolutional neural network [18] and scVAED uses a variational autoencoder (VAE) [19].

In this work, we focus on the scVAED framework that combines variational autoencoder with a diffusion processes [19]. We adopt their architecture as a representative LDM approach. However, not with the goal of generating data, but to assess whether diffusion-based refinement of learned embeddings can improve clustering quality and, indirectly, the reliability of DE-based cell type identification.

1.5 Overview of thesis structure and objectives

Accordingly, this thesis is organized into two parts. In Part 1, we evaluate how increasing transcriptional homogeneity affects the performance of standard DE analysis. By carefully sampling and subsetting from a multi-tissue atlas, we construct simulated samples of varying transcriptional homogeneity. We then evaluate the performance of a DE in these samples by the resulting marker genes biological significance in a GSEA.

In Part 2, we investigate whether alternative embedding strategies can improve cell clustering in both heterogeneous and homogeneous settings, and thereby help accommodate the limitations of DE observed in Part 1. More specifically, we will compare log-normalised highly variable gene counts(HVG), linear embeddings (PCA), nonlinear latent representations learned by variational autoencoders (VAEs), and diffusion-based refinements on VAE embeddings (LDM). The goal of the analysis is to evaluate whether refined latent representations preserve fine-grained biological structure more effectively than conventional approaches, leading to improved performance in downstream tasks such as clustering and label transfer.

Together, these two parts address a central question: can DE analysis be enhanced by performing clustering on refined embeddings produced by nonlinear methods (VAEs) and further improved through diffusion-based refinement (LDM)? Part 1 establishes the challenge—that DE performance degrades in homogeneous settings—while Part 2 explores whether advanced embedding methods can mitigate this limitation by producing more biologically meaningful cell groupings.

2 Data

For this thesis we used four datasets spanning a gradient of transcriptional heterogeneity. A summary of the key data characteristics of these datasets can be seen in table 1

2.1 DISCO Atlas:

To obtain a large and well-annotated dataset, we used 22 organ-specific scRNA-seq atlases from the Deeply Integrated Single-Cell Omics (DISCO) database [21]. DISCO aggregates publicly available scRNA-seq datasets into a harmonized framework and applies a standardized upstream processing pipeline to improve comparability across studies.

In brief, raw sequencing data are processed using a standardized preprocessing procedure, where all data are mapped to a common human reference genome, and subjected to cell-level quality control based on mitochondrial transcript proportions and feature counts. These procedures are applied uniformly across datasets to make data integration easier and reduce batch effects in downstream analyses [21]. Cell type labels are harmonized by the authors using the reference-based annotation tool CELLiD [21].

We selected 22 individual scRNA-seq datasets from DISCO, representing diverse tissue types (listed in Appendix A). To balance representation across tissues, we randomly sampled 20,000 cells from each dataset. The resulting collection comprised a total of 440,000 cells, 33,538 genes, and 328 unique annotated cell types.

2.2 Novo Nordisk Atlas:

To evaluate our method on heterogeneous data, we used an internal Novo Nordisk multi-tissue atlas composed of multiple datasets from both in-house experiments and public sources. Before analysis, and to avoid breaching confidentiality, the dataset was filtered to include only publicly sourced data. The full dataset consists of 731,407 cells and 40,405 genes. For the analysis, we subsampled the data to a maximum of 1,000 cells per cell type, yielding a final dataset of 27,242 cells across 32 different cell types (There were not 1,000 of each cell type originally). The dataset consist of data sourced from 12 different datasets, additional information about these datasets can be seen in appendix B. The dataset will be referred to as `allCT_1.0k` throughout the thesis.

2.3 In vitro β -cell differentiation (Veres et al., 2019):

As a biologically more homogeneous and therefore more challenging benchmark, we used the in vitro β -cell differentiation dataset from Veres et al. [22]. The study systematically profiled the developmental progression of stem-cell-derived pancreatic lineages. They followed a protocol that targeted beta cell differentiation.

We focused on Stage 6 of the protocol, representing the final maturation phase. The dataset contains 38,494 cells and 16,039 genes and is annotated at two resolutions:

- **Cluster:** broad cell-type categories obtained via an initial Leiden clustering.
- **Subcluster:** finer-resolution states obtained by reclustering within each broad cluster.

Annotations were assigned by the authors using canonical markers. The coarse annotation comprises 8 major cell types, while the fine-grained annotation consists of 17 subtypes. To achieve a balanced class representation, we subsampled to at most 1,000 cells per label, resulting in two derived datasets:

- **Coarse-label dataset:** 8 cell types, 7,475 cells.
- **Fine-label dataset:** 17 cell types, 12,680 cells.

These datasets provide a controlled setting for evaluating whether embedding methods capture subtle transcriptional differences within closely related endocrine lineages, they will be referred to as `GSE114412_cluster` and `GSE114412_subcluster` throughout this thesis.

2.4 DISCO subset

To specifically evaluate method performance in Part 2 under highly homogeneous conditions—the central challenge identified in Part 1—we constructed an additional test case from the fifth grouping category created in Part 1 of the thesis, named “Very similar cell types” (Group 5) (see section 3.2 for details). This subset was constructed by sampling cells from $N = 7$ cell types with a mean pairwise correlation $\rho \geq 0.9$. The resulting subset, which will be referred to as `group_5_subset_76_ncells_7` throughout the remainder of this thesis, contains a total of 12,597 cells, and the cell types exhibit a mean pairwise correlation of $\rho = 0.928$.

Table 1: This table summarizes the information about the datasets used in this the thesis. The “*Heterogeneity*” column was our expectations to the datasets heterogeneity. The “*Part*” column indicates which parts of the thesis is using the dataset

Dataset	Cells	Genes	Types	Heterogeneity	Part
DISCO atlas	440,000	33,538	328	High	1
Novo Nordisk atlas (<code>allCT_1.0k</code>)	27,242	40,405	32	High	2
Veres et al., 2019 (<code>GSE114412_cluster</code>)	7,475	16,039	8	Medium	2
Veres et al., 2019 (<code>GSE114412_subcluster</code>)	12,680	16,039	17	Low	2
DISCO subset (<code>group_5_subset_76_ncells_7</code>)	12,597	33,538	7	Very low	1, 2

3 Part 1: Differential Expression Analysis in Homogeneous Samples

3.1 Methods — Part 1

The goal of Part 1 was to test the hypothesis that standard Differential Expression (DE) analysis on homogeneous samples failed. To do this we created subsets of varying homogeneity from the DISCO atlas. Each subset was then grouped by cell type label and a Wilcoxon Rank Sum Test [23] was performed between the cell types. The quality of the DE analysis was then evaluated using the Differential Expressed Genes (DEG) which were used in a gene set enrichment analysis (GSEA). The DE analysis was deemed successful if the associated DEG of a cluster was effective canonical markers of the cell type and returned the correct label in the GSEA.

3.1.1 Preprocessing and normalization

Starting from the gene–cell count matrices collected from the DISCO database, all preprocessing and analysis in Part 1 was performed using Seurat. The datasets were merged and normalized using the `Seurat::NormalizeData` function with default parameters [4]. Specifically, total counts for each cell were scaled to a fixed value of 10,000, followed by natural log-transformation with a pseudocount of 1.

This normalization procedure corrects for differences in sequencing depth between cells while stabilizing variance across varying expression range. No additional preprocessing steps were applied after normalization. This choice was intentional, as the aim was to evaluate the behavior of standard differential expression based marker gene discovery, which is commonly performed on normalized counts [9].

3.1.2 Sampling Subsets of Varying Homogeneity

To construct groups of varying transcriptional homogeneity, we sampled subsets of cell types from the DISCO atlas. Group 1 and 2 were motivated biologically, and consisted of subsets of a single cell type, from either one tissue (Group 1) or multiple tissues (Group 2). The remaining 7 groups (Group 3-9), were constructed from computed similarities and represented increasing levels of heterogeneity.

We quantified similarity between all pairs of cell types, by computing the Pearson correlation between the average expression profiles of all cell types. Here, average expression profile means a vector with the mean gene expressions for all cells of a given cell type. Thus we would have a mean expression

profile for each cell type in the data. We let $\rho = cor_{x,y}$ where x and y denote the mean expression vectors of two cell types. Positively correlated pairs are interpreted as transcriptionally similar, whereas weakly or negatively correlated pairs represent increasingly diverse cell types.

The subset categories used for sampling were defined directly from these correlations (Table 2), and constructed by selecting cell types within specific correlation intervals.

For each subset category, sampling proceeded as follows:

1. First, a random cell type was selected.
2. Then all cell types whose correlations fell within the similarity interval associated with the group category were selected as eligible cell types.
3. From the eligible cell types, n cell types were sampled uniformly at random to form a subset of cell types.

This procedure was repeated five times for each group category and for group categories allowing variable size ($2 - N$) we sampled five groups for each $n \in \{2 : N\}$, and we let $N = 20$.

Group	Description	Similarity range
No similarity definition		
1	One cell type from a single tissue	—
2	One cell type sampled across multiple tissues	—
Very similar cell types		
3	Two very similar cell types	$\rho \geq 0.9$
4	Three very similar cell types	$\rho \geq 0.9$
5	N very similar cell types	$\rho \geq 0.9$
Slightly more diverse cell types		
6	$2-N$ slightly more diverse cell types	$0.7 \leq \rho < 0.9$
Diverse cell types		
7	$2-N$ diverse cell types	$0.40 \leq \rho < 0.7$
Highly diverse cell types		
8	$2-N$ highly diverse cell types	$0 \leq \rho < 0.40$
Random composition		
9	$2-N$ random cell types (bootstrapped)	—

Table 2: Group definition and their similarity range.

3.1.3 Differential Expression analysis and Evaluation

To assess how transcriptional homogeneity influences differential expression (DE) performance, we conducted DE analysis on each sampled subset group and evaluated the biological relevance of the resulting marker genes.

DE testing was performed between cell-type-defined clusters within each subset ¹ using the Wilcoxon rank-sum test implemented in Seurat’s FindAllMarkers() [23, 25]. This yielded a set of positively upregulated marker genes for every cell-type cluster in each sampled group. All parameter settings can be found in Appendix E.

To evaluate how well these markers reflected true cell-type identity, we applied gene set enrichment analysis (GSEA) using the DISCO toolkit (DiscoToolkit::scEnrichment()) [21]. The method compares

¹For Group 1, there were only one cell type in the subsets. Preceding DE analysis a $k-nn$ graph and subsequent Leiden clustering[24] was performed. The resulting clusters were used for the DE analysis.

the DE-derived marker genes and their log fold changes to curated reference marker gene sets across DISCO atlases and computes odds ratios for enrichment. For each query, the tool returns the highest-ranking reference cell types along with associated p-values.

From these results, we derived three metrics indicating how accurately the DE analysis recovered the expected cell type (i.e., the annotation used during sampling):

1. **Top-1 match:** whether the correct cell type appears as the top enriched cell type.
2. **Top-5 match:** whether it appears within the top five enriched cell types.
3. **No match outcome:** whether it appears outside of the top five enriched cell types.

Together, these evaluation metrics quantify how reliably DE analysis can identify biologically meaningful marker signatures as the similarity structure of the sampled subsets becomes increasingly heterogeneous.

3.2 Results — Part 1

Sampling To ensure that the sampled subsets captured the intended levels of transcriptional similarity, we examined the mean within group correlations by group category (see figure 2). Very similar subsets (Group 3-5) showed high mean correlations ($\rho \geq 0.9$, while the "slightly diverse" (Group 6), "diverse" (group 7) and "highly diverse" (Group 8), showed progressively lower correlation. Random groups showed correlations around the overall mean of the pairwise correlations. The biological motivated Group 2 showed correlations above 0.8. This supports pairwise correlation as a suitable metric for evaluating within-sample homogeneity, as expected high homogeneity of group 2 was reflected in the similarity score. Overall, these results confirm that the sampling procedure successfully generated subsets spanning different similarity levels.

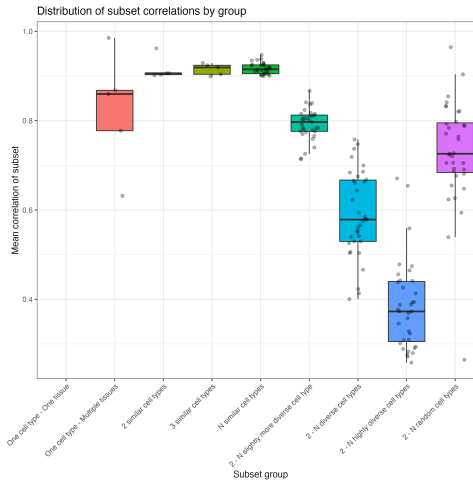


Figure 2: Mean within-group correlation for each sampled subset, stratified by subset category. Each dot represents a subset.

Differential Expression Performance Across Similarity Levels DE performance was evaluated using GSEA on DE-derived marker genes, quantified by Top-1 Accuracy and Top-5 accuracy. The results of the analysis is visualised in figure 3, which shows the proportion of cell types that were correctly recovered by GSEA across the different subset groups. Performance was lowest in the most homogeneous groups: both “One cell type” condition (Group 1 and 2) and the “similar cell type” conditions (group 3-5) showed a high proportion of No Match outcomes and low Top-1 accuracy. As the subsets became more diverse, the proportion of correct Top-1 matches increased, reaching the highest performance in the “diverse” and “highly diverse” groups (Group 6-8). Random groups showed intermediate performance. Overall, DE analysis was least reliable in homogeneous subsets, consistent with the expectation that homogeneous samples would lose discriminative signal.

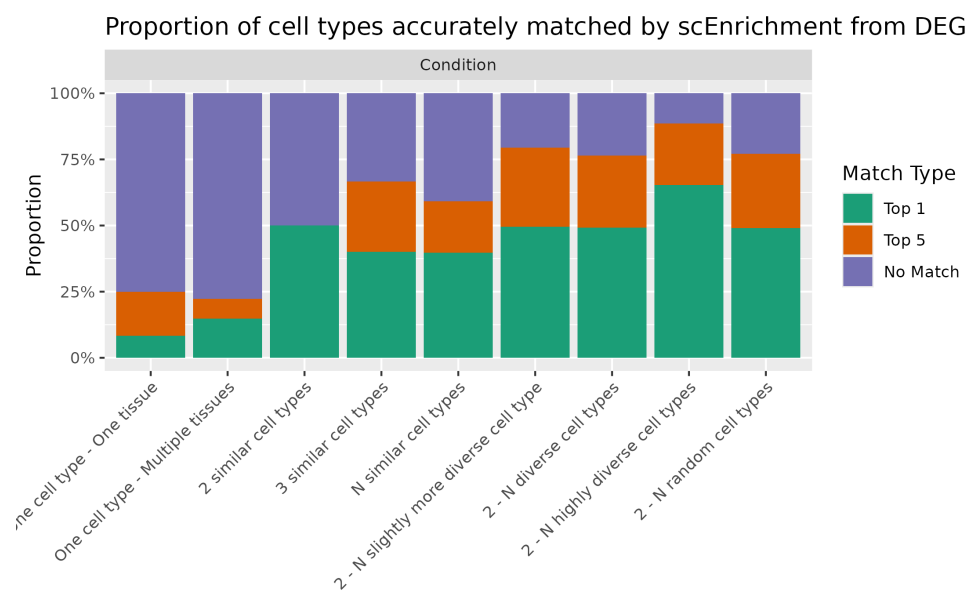


Figure 3: Stacked bar plot showing Top-1 and Top-5 GSEA accuracy for each subset group category.

4 Part 2: Evaluating Latent Representations of scRNA data

4.1 Methods — Part 2

The goal of part 2 was to assess whether embeddings derived from a Variational Autoencoder (VAE) followed by denoising with a latent diffusion model (LDM) could provide biologically more relevant representations of different cell types compared to standard dimensionality reduction methods such as PCA. We compared four embedding approaches—highly variable genes (HVG), PCA, VAE, and LDM—using two complementary metrics: the cell-type Local Inverse Simpson Index (cLISI) score to measure local neighborhood homogeneity, and label transfer accuracy using logistic regression to assess global class separability, see figure 4 for an overview. It is important to note that the ground-truth cell-type labels used for evaluation were originally derived from PCA-based clustering workflows, which introduces a potential bias favoring PCA-based embeddings. These annotations nevertheless represent the closest available approximation to biological ground truth. Optimal hyperparameters (latent dimensions and denoising steps) were selected via grid search maximizing cLISI, and stability was evaluated through 10 independent runs per dataset with different random seeds.

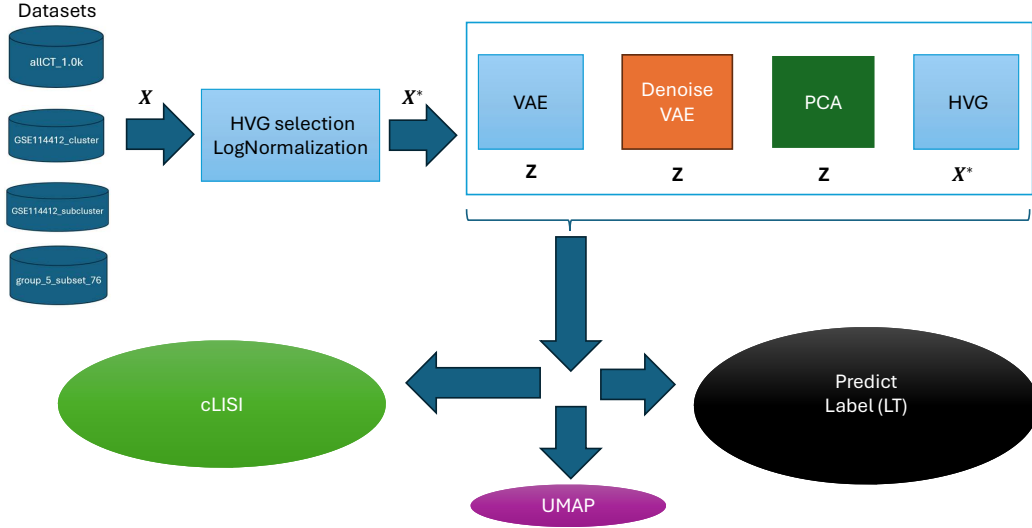


Figure 4: A schematic overview of the data, methods and evaluation metrics used in part 2 of this thesis. X is the raw data frame, X^* is the preprocessed dataframe, also referred to as the HVG representation. Z symbolizes a lower dimensional representation of X^* .

4.2 Data

The four datasets used in this analysis are the same as those described in the Data section. We expect `allCT_1.0k` to represent a highly heterogeneous dataset, while the remaining three datasets (`GSE114412_cluster`, `GSE114549_cluster`, and `group_5_subset_76`) are expected to be more homogeneous.

Data preprocessing

4.2.1 Normalization and log-transformation

In contrast to part 1, which utilized the R package Seurat[4], part 2 employs Python’s single-cell analysis library Scanpy[5] for all general scRNA-seq preprocessing. Raw count data were first normalized and log-transformed. Library size normalization was performed by scaling each cell’s counts to a total count of 10,000, followed by natural log-transformation with a pseudocount of 1 using $\ln(x + 1)$. This preprocessing reduces technical variation due to sequencing depth differences and stabilizes variance across the range of expression. Normalization and log-transformation were implemented using Scanpy’s `sc.pp.normalize_total` and `sc.pp.log1p` functions [5].

4.2.2 Highly variable gene identification

Following normalization of the training set, highly variable genes (HVGs) were identified using Scanpy’s implementation: `scanpy.pl.highly_variable_genes` following the method described by Satija et al. [26]. Selecting HVGs is standard practice in scRNA-seq analysis, as it reduces computational burden while retaining genes that carry biological signal [27], and it is also used as input in scVAEDer [19]. We selected the top 2,000 HVGs, a number chosen within the range of best practices in the field[28]. HVG expression values serve as a simple baseline representation against which we compare learned embeddings. For datasets containing multiple samples or batches, HVGs were identified within each batch separately using the `batch_key` parameter to avoid selection of batch-specific genes. This approach identifies genes with high variability relative to their mean expression by:

1. Binning genes by mean expression
2. Computing normalized dispersion (variance/mean) within each bin
3. Ranking genes by their normalized dispersion (or across batches when applicable)

The HVGs were selected based on the training data, and all three data splits (training, validation, and test) were subsequently subset to these 2000 HVGs to ensure consistency across splits.

4.3 Embedding methods

4.3.1 PCA implementation

Principal Component Analysis (PCA) was performed using scikit-learn’s `PCA` class [29], which implements singular value decomposition (SVD). The number of principal components used varies in the range mentioned in section w refsec:gridsearch.

4.3.2 Variational autoencoder

A Variational Autoencoder (VAE) is a generative model that learns a probabilistic mapping between high-dimensional data x and a low-dimensional latent representation z [30]. Unlike standard autoencoders that learn deterministic encodings, a VAE learns the parameters of a distribution over the latent space by first encoding x and decoding z .

The **encoder** maps input data to latent distribution parameters. For a Gaussian latent distribution:

$$q(z|x) = \mathcal{N}(z | \mu_z(x), \sigma_z^2(x)) \quad (1)$$

where the encoder outputs μ_z and $\log \sigma_z^2$. A latent sample is drawn using the reparameterization trick: $z = \mu_z + \epsilon \cdot \sigma_z$ where $\epsilon \sim \mathcal{N}(0, I)$.

The **decoder** models the likelihood $p(x|z)$, reconstructing data from the latent code.

VAE Loss Function The VAE loss combines two components:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \beta \cdot \mathcal{L}_{\text{KL}} \quad (2)$$

where $\beta \geq 0$ is a weighting hyperparameter that controls the trade-off between reconstruction fidelity and latent space regularization.

Reconstruction Loss measures how well the decoder reconstructs input data:

$$\mathcal{L}_{\text{recon}} = -\mathbb{E}_{q(z|x)}[\log p(x|z)] \quad (3)$$

For our log-normalized, highly variable gene (HVG) preprocessed scRNA-seq data, we use a Gaussian likelihood with mean squared error (MSE). The decoder outputs μ_x , and the reconstruction loss is:

$$\mathcal{L}_{\text{recon}} = \sum_{i,j} (x_{ij} - \mu_{x,ij})^2 \quad (4)$$

This assumes that after log-normalization, gene expression values are approximately normally distributed with relatively homogeneous variance across genes.

KL Divergence regularizes the latent space to match a prior $p(z) = \mathcal{N}(0, I)$:

$$\mathcal{L}_{\text{KL}} = \text{KL}[q(z|x)||p(z)] = \frac{1}{2} \sum_{k=1}^K (\mu_{z,k}^2 + \sigma_{z,k}^2 - \log \sigma_{z,k}^2 - 1) \quad (5)$$

Where K is the dimensionality of the latent space z .

Complete VAE Loss The complete VAE loss is:

$$\mathcal{L}_{\text{VAE}} = \sum_{i,j} (x_{ij} - \mu_{x,ij})^2 + \beta \sum_i \text{KL}[q(z_i|x_i)||p(z)] \quad (6)$$

where the first term encourages faithful reconstruction and the second term regularizes the latent space to be smooth and continuous, enabling meaningful interpolation and generation.

4.3.3 Diffusion Denoising Probabilistic Model(DDPM)

A diffusion model consist of a forward and backward process, and is overall modeled as a first order markov chain, which means that the probability of transitioning to the next state only depends on the current state i.e $p(z_t|z_{t+1})$ [31].

The forward process, also known as the diffusion, progressively adds noise to the input data z_0 over a sequence of discrete timesteps $t \in \{1, \dots, T\}$, where T denotes the total number of diffusion steps. The noise comes from a probability distribution where we in our case use a Gaussian distribution. Using the Gaussian distribution provides us with term (7) that enables us to calculate any noised version of z_0 .

$$q(z_t|z_0) = \mathcal{N}(z_t; \sqrt{\bar{\alpha}_t}z_0, (1 - \bar{\alpha}_t)I) \quad (7)$$

where $\bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i)$ is dependent on β_t and defines the noise schedule for each timestep. (7) can be reparameterized as:

$$z_t(z_0, \epsilon) = \sqrt{\bar{\alpha}_t}z_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon \quad (8)$$

for $\epsilon \sim \mathcal{N}(\mathbf{0}, I)$

The reverse process is where we learn to iteratively denoise towards z_0 from noisy versions $z_{1,2,\dots,T}$. In practice, we first sample a timestep t and calculate the noisy data z_t using Equation 8. We then train a neural network to predict the noise ϵ that was added to create z_t . The network's prediction is denoted as ϵ_θ , where θ represents the network parameters. These parameters are optimized by minimizing the loss function:

$$\mathcal{L}_{\text{diffusion}} = \mathbb{E}_{t,z_0,\epsilon} \|\epsilon - \epsilon_\theta(z_t, t)\|^2 \quad (9)$$

4.3.4 Refining VAE Latents through Partial DDPM Denoising

To obtain refined latent representations, we employ a partial denoising strategy that leverages the trained diffusion model. Given a clean VAE-encoded latent z_0^{VAE} and a chosen denoising depth $t_{\text{denoise}} \in [1, T]$, the refinement process consists of two stages:

Forward noising: We first add controlled Gaussian noise to the clean latent by applying the forward diffusion process (Equation 8) at timestep t_{denoise} :

$$z_{t_{\text{denoise}}} = \sqrt{\bar{\alpha}_{t_{\text{denoise}}}} z_0^{\text{VAE}} + \sqrt{1 - \bar{\alpha}_{t_{\text{denoise}}}} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (10)$$

Reverse denoising: Starting from $z_{t_{\text{denoise}}}$, we iteratively apply the learned reverse process to denoise back to the clean state. At each timestep i from t_{denoise} down to 0, we compute:

$$z_{i-1} = \frac{1}{\sqrt{\alpha_i}} \left(z_i - \frac{\beta_i}{\sqrt{1 - \bar{\alpha}_i}} \epsilon_{\theta}(z_i, i) \right) + \sigma_i \epsilon^*, \quad \epsilon^* \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (11)$$

where $\sigma_i = \sqrt{\beta_i \frac{1 - \bar{\alpha}_{i-1}}{1 - \bar{\alpha}_i}}$ is the posterior variance, and ϵ_{θ} is the trained noise prediction network. For the final step ($i = 0$), no noise is added.

The resulting denoised latent z_0^{LDM} serves as our refined representation. The parameter t_{denoise} controls the strength of refinement: smaller values apply gentler denoising, while larger values introduce more substantial modifications to the original VAE latent structure.

4.4 Model implementation and training process

Beyond the mathematical formulation of the VAE and the DDPM, several implementation and architectural design choices must be made. These include the neural network architectures used to parameterize the encoder $q_{\phi}(z|x)$, decoder $p_{\theta}(x|z)$, and noise prediction network $\epsilon_{\theta}(z_t, t)$, as well as diffusion-specific hyperparameters such as the number of diffusion steps T and the noise schedule $\{\beta_t\}_{t=1}^T$. For both the VAE and DDPM implementations used in this project, we adopted the architectures and hyperparameters from scVAEDer [19]. The model was implemented python using pytorch [32], see 7 for our implementations.

4.4.1 VAE implementation

In the VAE, separate neural networks parameterize the encoder $q_{\phi}(z|x)$ and decoder $p_{\theta}(x|z)$, where ϕ and θ denote the learned network weights. We adopted the architecture from scVAEDer [19]. The encoder outputs the latent distribution parameters μ_z and $\log \sigma_z^2$. It consists of three fully connected layers with dimensions $[2,000 \rightarrow 100 \xrightarrow{\mu_L} \log \sigma_L^2]$, where $L \in \{10, 20, 30, 40, 50\}$ was optimized via grid search. A latent representation z is then sampled from the learned distribution using the reparameterization trick, in which gaussian noise is combined with the encoder outputs μ_L and σ_L with $z_L = \mu_L + \epsilon \times \sigma_L$. The decoder maps the latent code z_L back to a reconstructed input $\hat{x}$, it consist of 2 fully connected layers with dimensions $[L \rightarrow 100 \rightarrow 2,000]$. A more detailed overview of the implementation choices for the VAE model can be seen in Appendix C.

4.4.2 Diffusion implementation

For the diffusion model, we adopted the architecture from scVAEDer. The noise prediction network $\epsilon_{\theta}(z_t, t)$ operates on the VAE latent space and takes both the noised latent code z_t and the diffusion timestep t as inputs. The timestep t is embedded using a learned time embedding and incorporated into the network to condition the denoising process on the current diffusion stage. The network consists of 4 fully connected layers with dimensions $[L \rightarrow 512 \rightarrow 512 \rightarrow 512 \rightarrow L]$ where $L \in \{10, 20, 30, 40, 50\}$ was optimized via grid search. A detailed overview of the implementation choices for the diffusion model can be seen in Appendix C.

4.4.3 Training procedure

All models were trained using the same data split: 70% train, 20% validation, and 10% test. Highly variable genes (HVGs) were identified on the training set only, and all three splits were subsetted to the 2000 HVGs. These HVGs served both as a baseline representation and as the input to all embedding methods (PCA, VAE, and LDM).

The VAE was trained on the training set and monitored on the validation set using early stopping based on the validation loss. After training, the final encoder was used to embed the full training set into its latent space z . These training-set latent embeddings were then used to train the denoising

network of the diffusion model. The denoiser was optimized to predict the Gaussian noise added during the forward diffusion process by minimizing the mean squared error, using the AdamW optimizer. Architectural and hyperparameter details for the diffusion model are provided in Table 5.

Once trained, the VAE encoder and diffusion model were applied sequentially to the test set to obtain both VAE and diffusion-refined embeddings for downstream evaluation.

Finally, PCA was fitted using only the training set. The learned loadings were then applied to project the test set into the space spanned by the principal components, allowing for comparison with the VAE and diffusion-based embeddings.

4.5 Model Evaluation

To evaluate the quality of the latent representations produced by the different embedding methods, we used two complementary metrics. Both metrics quantify how well the embeddings preserve biological variation associated with cell type. One capture local neighborhood consistency while the other capture global class separability.

cLISI: cell-type Local Inverse Simpson Index: The cLISI score quantifies how homogeneous the local neighborhoods are with respect to cell-type labels. For each cell, a k -nearest neighbor graph is constructed (with $k = 90$), and the inverse Simpson index is computed based on the distribution of labels in its neighborhood. We used the implementation from `scanpy` [5], in which a cLISI score close to 1 indicates that nearby cells tend to share the same annotation.

Label Transfer Accuracy: As a complementary metric, we evaluated how well each embedding supports supervised classification. A multiclass logistic regression classifier from `SciPy` [33] was trained on the training set embeddings for each method (see appendix D for details). The trained classifier was then used to predict cell type labels on the test set embeddings. Accuracy score was computed on the predictions versus groundtruth labels. High performance of the classifier indicates that the embedding method successfully preserves meaningful biological structure that can separate cell types in latent space. u

Labels: Both cLISI and label transfer accuracy rely on ground-truth cell-type annotations. For the two beta-cell differentiation datasets, cell types were annotated through expert-guided marker gene analysis of clusters derived in PCA space. This workflow represents standard annotation practice in scRNA-seq analysis [9]. It is therefore reasonable to assume that most annotations in the DISCO and allCT_1.0k atlases originate from comparable PCA-based clustering and marker interpretation procedures. As a consequence, the evaluation labels are not independent of PCA, which may introduce a bias favoring PCA-based embeddings in downstream comparisons. However, these annotations represent the closest available approximation to biological ground truth and are widely accepted in the field. Given that experimentally validated cell-type labels at this scale are not easily obtained, they provide a practical basis for evaluating the embedding quality with the metrics describe above.

4.5.1 Grid search

To determine the optimal latent dimensionality for PCA and the VAE, and the optimal number of denoising steps for the LDM, we performed a grid search over the following hyperparameters. The latent dimension was varied over five values: $L \in \{10, 20, 30, 40, 50\}$, and the number of diffuse-denoise steps was varied over seven values, $t \in \{0, 5, 10, 20, 40, 80, 160\}$, where $t = 0$ corresponds to using the raw VAE latent embedding without diffusion refinement, this range was chosen on the basis of what is used in image refinement [16]. The grid search was conducted independently for each dataset, and the optimal combination of L and t was selected based on the model that maximized the cLISI score.

4.5.2 Repeated Runs with Different Random Seeds

To assess the stability of each embedding method and obtain more reliable estimates of performance, we performed repeated runs using different random seeds. For each dataset the latent dimensions L and number of diffusion steps, t , were fixed to the values selected in the grid search. Each run used a

new random seed, which affected the train/validation/test split as well as the parameter initialization of the VAE, diffusion model, PCA, and logistic regression classifier. In total, we performed 10 independent runs per method per dataset.

4.5.3 Statistical Testing

To test statistical difference between the methods, we performed a pairwise Wilcoxon signed-rank test within each dataset and for each metric. As we have four methods, we get six pairwise comparisons per dataset and a total of 24 comparisons per metric. Since all methods were evaluated by running different random seeds with identical data splits for each seed, we treated comparisons as paired.

The Wilcoxon signed-rank test was chosen, because it takes paired observations into account and does not assume normality of the underlying distribution, which is important given the limited number of repeated runs (10 per method per dataset). The test evaluates whether the median of the pairwise differences between two methods deviates significantly from zero [34], thereby assessing whether one method consistently outperforms another across repeated runs.

To account for multiple hypothesis testing within each dataset we adjusted p-values with the Benjamini-Hocberg false discovery rate (FDR) correction. FDR correction controls the expected proportion of false positives, making it well suited for multiple comparisons. Adjusted p-values below 0.05 were considered statistically significant.

4.5.4 Dimensionality Reduction Visualization

We employed Uniform Manifold Approximation and Projection (UMAP) [35] to visualize high-dimensional representations in two dimensions. UMAP was applied to raw log-normalized data, PCA embeddings, VAE latent representations, and denoised VAE latent representations. All UMAP embeddings used consistent hyperparameters: 15 nearest neighbors, minimum distance of 0.1, Euclidean metric, spectral initialization, and fixed random seed for reproducibility. UMAP was implemented using the `umap-learn` Python package [35].

Results Part 2

4.6 cLISI Score Analysis

Grid search: The grid search across latent dimension sizes and denoising steps revealed varying cLISI scores ranging from 0.8211 to 0.9934 between datasets and methods datasets (Figure 5). PCA consistently demonstrated the best performance, while highly variable genes (HVG) underperformed compared to all other methods.

For the most heterogeneous dataset (allCT_1.0k), all methods and hyperparameters achieved similarly high cLISI scores. The two late-stage beta cell differentiation datasets (GSE114412_cluster and GSE114412_subcluster) exhibited greater variation across latent dimension sizes and denoising steps. In GSE114412_cluster (Figure 5), dim 40 (red) and dim 50 (purple) showed slight increases in cLISI scores as denoising steps increased to 40 and 20, respectively.

The synthetically created DISCO dataset subset (group_5_subset76) showed that increasing denoising steps generally decreased cLISI scores. All methods achieved higher cLISI scores on this synthetic dataset compared to the beta cell differentiation datasets.

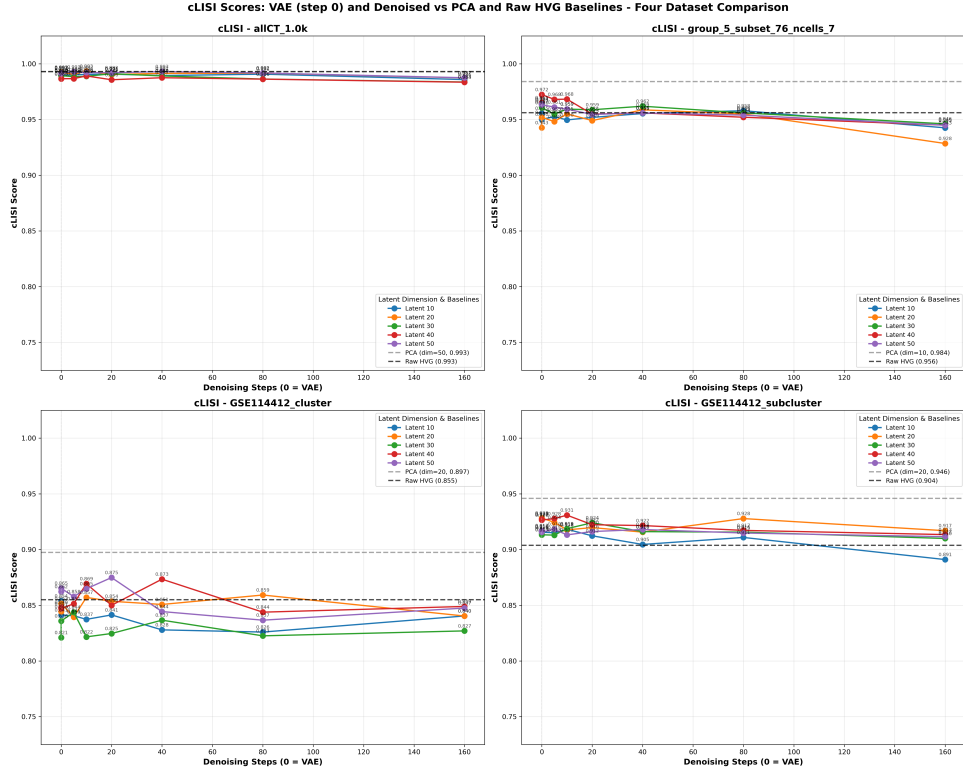


Figure 5: Grid search results comparing cLISI scores across latent dimensions and denoising steps for four datasets. Each subplot shows cLISI scores as a function of denoising steps (x-axis) for multiple VAE latent dimensions (colored lines). Point at denoising step 0 represents the baseline VAE embedding (no denoising), while subsequent points show progressive denoising refinement. Horizontal dashed gray line shows best PCA score with corresponding dimensionality, black dashed line shows raw HVG baseline. **Top row:** allCT_1.0k (left) and group_5_subset_76 (right). **Bottom row:** GSE114412_cluster (left) and GSE114412_subcluster (right). All subplots share the same y-axis scale (0.75–1.0) for direct comparison. PCA consistently outperforms VAE-based methods across all datasets and hyperparameter configurations. Denoising shows minimal improvement over baseline VAE embeddings, with most trajectories remaining flat or showing marginal changes across denoising steps.

Repeated runs: Ten consecutive runs were performed using the best-scoring hyperparameters for each method. The best-scoring hyperparameters, average cLISI scores and standard deviations are shown in Figure 6. The allCT_1.0k dataset showed highly consistent cLISI scores across runs with minimal variation between methods. For the GSE114412_cluster dataset, PCA(cLISI: 0.8927 ± 0.0185 | $p < 0.05$) demonstrated a significantly better performance, followed by VAE(cLISI: 0.8570 ± 0.0123) and the denoised embeddings(LDM)(cLISI: 0.8566 ± 0.011), with HVG(cLISI: 0.8382) consistently showing the lowest scores. The GSE114412_subcluster dataset showed similar patterns. The relative ranking of the methods showed a similar pattern across data sets. VAE and LDM did not perform significantly different in any of the datasets and PCA always performed equally well or better, with the difference being significant for group_5_subset_76, GSE114412_cluster and GSE114412_subcluster. A complete overview of the Wilcoxon sum rank test on the distributions of cLISI scores can be seen in Appendix F.

4.7 Label Transfer Accuracy

Grid Search: The grid search results using transfer accuracy as the performance metric revealed mixed outcomes across datasets (Figure 8). For both beta cell differentiation datasets (GSE114412_cluster and GSE114412_subcluster), HVG and PCA generally outperformed all VAE and LDM for all hyperparameter configurations.

For the group_5_subset_76 and allCT_1.0k datasets, grid search showed that HVG performed better than than all other methods, however the results also suggested that embedding using VAE and VAE

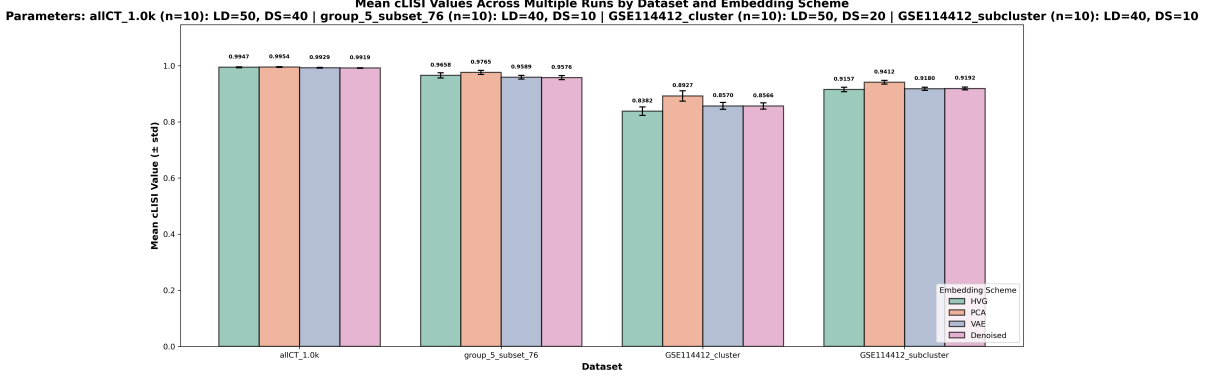


Figure 6: Mean cLISI scores ($\pm$ standard deviation) across 10 runs for four embedding methods and four datasets. PCA consistently outperforms VAE-based methods across all datasets. Error bars represent variability across independent training runs. Dataset-specific hyperparameters (latent dimension and denoising steps) were optimized via grid search.

+ LDM could have an improvement over PCA. Increasing the number of denoising steps did not consistently improve label transfer performance across datasets (Figure 8).

Repeated Runs: The ten consecutive runs (Figure 9) showed dataset-specific patterns. For allCT_1.0k, HVG achieved significantly higher label transfer accuracy than all other methods ($p = 0.0023$). However, both VAE (accuracy: $0.8975 \pm 0.0089 | p < 0.05$) and LDM (accuracy: $0.8880 \pm 0.0074 | p < 0.05$) significantly outperformed PCA (accuracy: 0.827 ± 0.0067). The group_5_subset.76 dataset showed that VAE (accuracy: $0.9515 \pm 0.0073 | p = 0.0059$) and LDM (accuracy: $0.9497 \pm 0.0078 | p < 0.05$) significantly outperformed both HVG (accuracy: 0.9490 ± 0.0069) and PCA (accuracy: 0.9309 ± 0.0084). For both beta cell differentiation datasets, PCA achieved significantly higher accuracy than all other methods (see Appendix F for a complete overview of all Wilcoxon signed-rank test results).

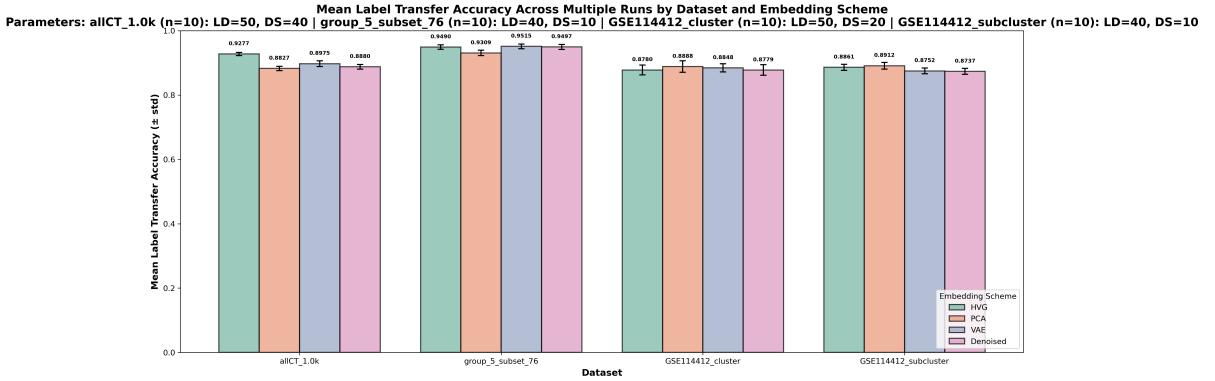


Figure 9: Mean label transfer classification accuracy ($\pm$ standard deviation) across 10 runs for four embedding methods and four datasets. Error bars represent variability across independent training runs. Dataset-specific hyperparameters were optimized via grid search.

4.8 cLISI vs Label Transfer Accuracy

cLISI scores and label transfer (LT) accuracy exhibited a positive correlation, indicating that better preservation of local cell type structure (high cLISI) generally corresponded to improved classification performance (high LT accuracy). This relationship is visually apparent in Figures 6 and 9, where higher cLISI scores consistently demonstrated higher LT accuracy.

The correlation between cLISI and LT accuracy is quantified in Figure 10, which reveals notable method-dependent patterns. For the HVG baseline, a strong positive correlation was observed ($r = 0.806$, $p = 3.54 \times 10^{-10}$), demonstrating that well-separated cell types in the original feature space

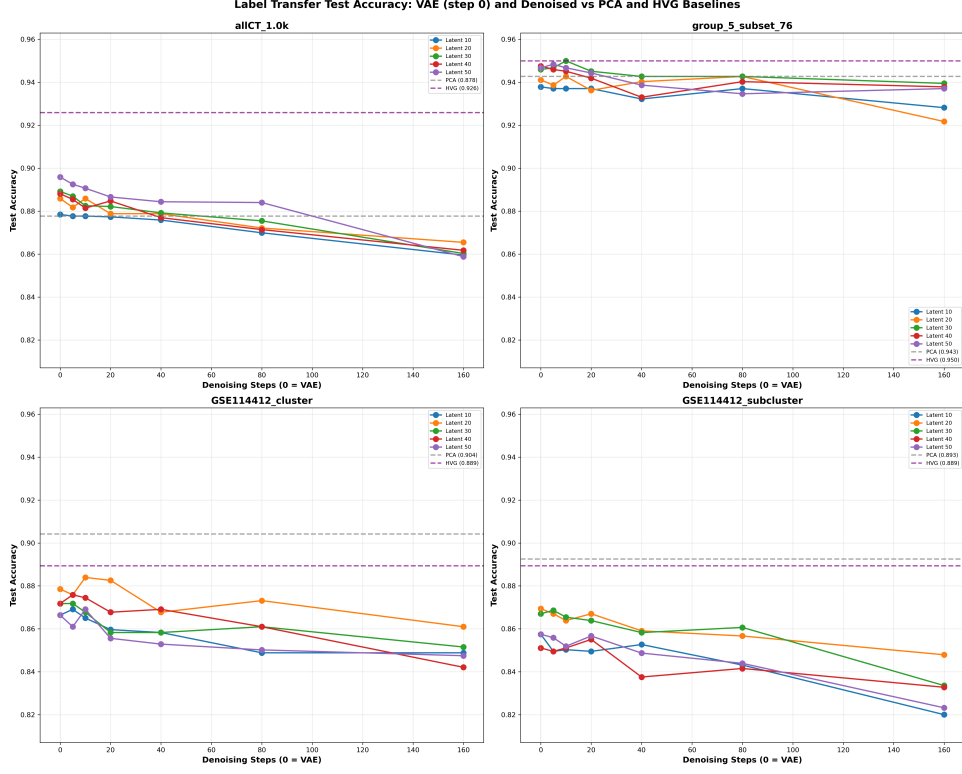


Figure 7: Label transfer accuracy across VAE latent dimensions and denoising steps for four datasets. Point at step 0 represents VAE baseline; subsequent points show denoising refinement. Dashed lines indicate PCA (gray) and HVG (purple) baselines.

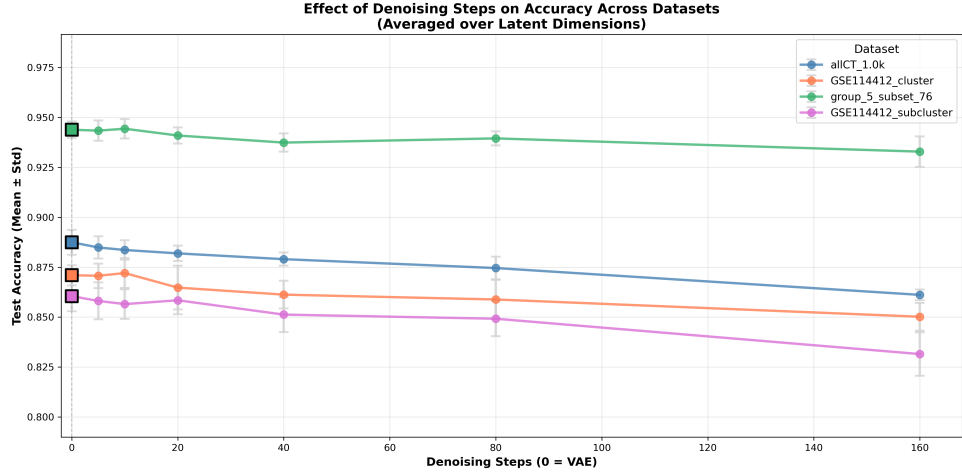


Figure 8: Effect of denoising steps on label transfer classification accuracy across datasets. Point at step 0 (square marker) represents baseline VAE; subsequent points show progressive LDM denoising. Trajectories averaged over all latent dimensions. Error bars indicate standard deviation. Denoising shows minimal effect across all datasets, with flat or slightly declining trajectories.

translate directly to accurate label transfer. In contrast, PCA exhibited a weak, non-significant correlation ($r = 0.248$, $p = 0.122$), suggesting that PCA’s dimensionality reduction may disrupt the relationship between local structure preservation and classification accuracy. VAE embeddings showed a moderate correlation ($r = 0.438$, $p = 4.75 \times 10^{-3}$), while LDM denoised embeddings demonstrated a similar moderate correlation ($r = 0.384$, $p = 0.0144$).

Across all methods, data points clustered by dataset identity, with `group_5_subset_76` consistently achieving the highest performance for label transfer accuracy (orange squares), followed by `allCT_1.0k` (blue circles) which consistently achieved highest cLISI scores. Finally, `GSE114412_subcluster` (red diamonds), and `GSE114412_cluster` (green triangles) scored consistently lower than the two other datasets. This dataset-specific clustering pattern held consistently across all four methods, indicating that inherent biological structure and data quality—rather than method choice—are the primary determinants of both cLISI and LT accuracy in our experiments.

It should be noted that both evaluation metrics exhibited a restricted range across all experiments. cLISI scores ranged from 0.82 to 0.99, while label transfer accuracy ranged from 0.85 to 0.96, representing a relatively narrow subset of each metric’s full scale. As a result, the observed correlations are derived from relatively small performance differences, and their magnitude should be interpreted with caution. Nevertheless, the consistency of the correlation patterns across methods and datasets, suggests that the relationship between local neighborhood structure (cLISI) and classification performance (Label Transfer Accuracy) remains informative within this constrained regime.

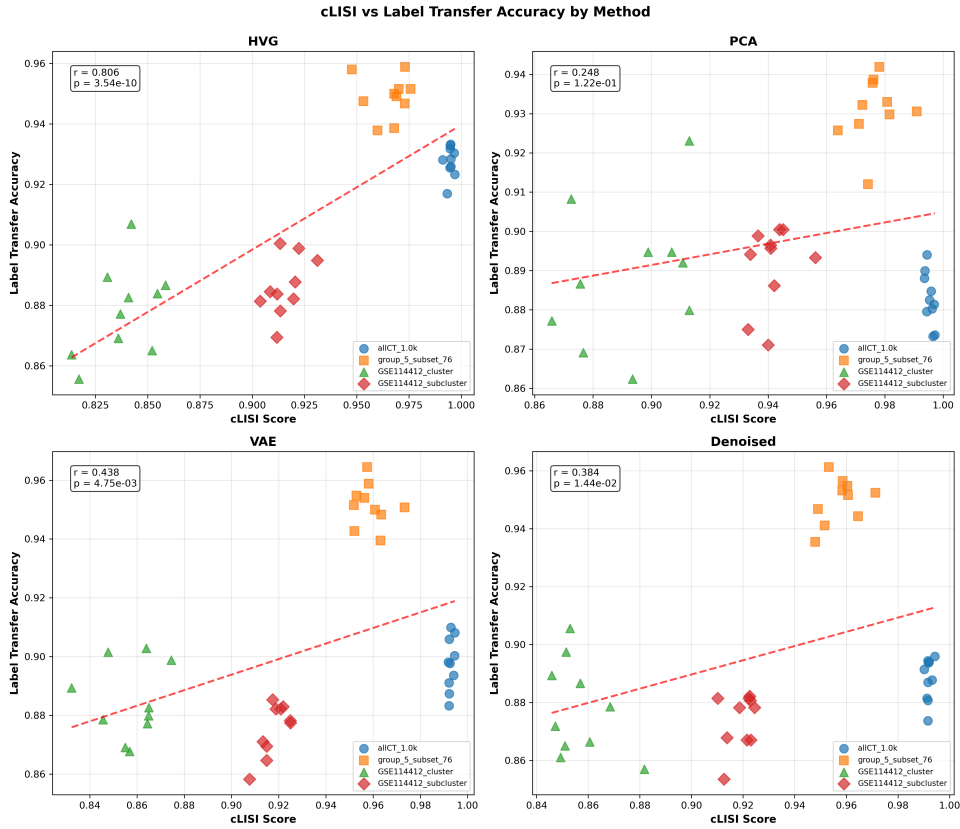


Figure 10: Correlation between cLISI scores and label transfer accuracy by method. Each subplot shows one embedding method with datasets represented by different colors/markers. Red dashed line: linear regression. HVG shows strong correlation ($r = 0.806$, $p = 3.54 \times 10^{-10}$), while PCA shows weak, non-significant correlation ($r = 0.248$, $p = 0.122$).

4.9 UMAP Visualization of Different Embedding Methods

UMAP projections were generated for HVG, PCA, VAE, and LDM representations across all four datasets using their respective optimal hyperparameters (Figure 11).

The HVG representation showed dispersed cell distributions, with allCT_1.0k exhibiting the highest degree of fragmentation. PCA representations demonstrated intermediate levels of cell population condensation across all datasets. VAE latent spaces displayed more cohesive and continuous cell population structures compared to linear methods. Denoised VAE representations showed similar spatial arrangements to non-denoised VAE embeddings, with subtle refinements in cluster boundaries most visible in the GSE114412 datasets.

The group_5_subset_76_ncells_7 dataset consistently showed well-defined cluster separation across all embedding methods. The allCT_1.0k dataset maintained greater cell population dispersion across all methods, while both GSE114412 datasets displayed intermediate characteristics with compact, distinct cell populations in VAE-based representations.

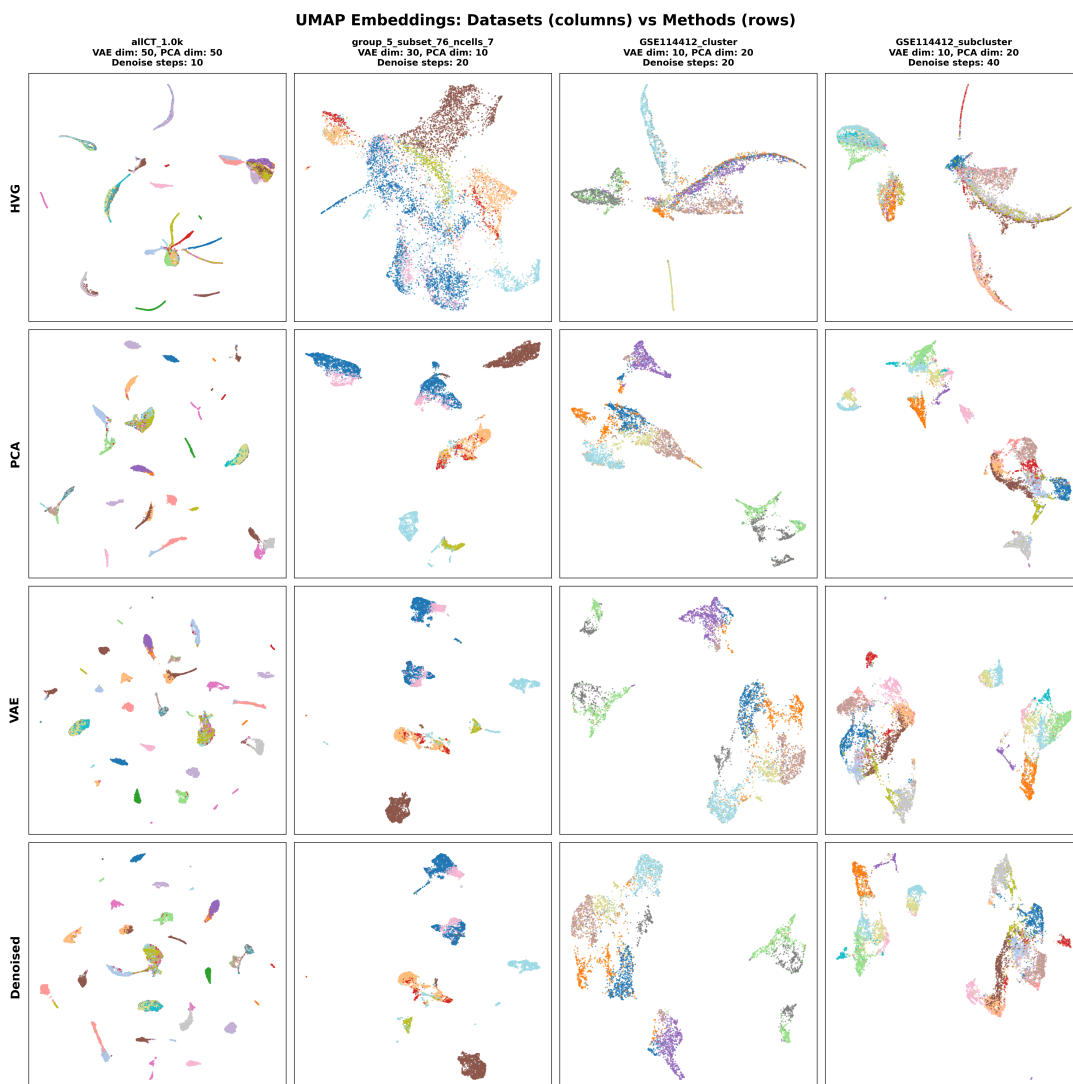


Figure 11: UMAP visualization of embeddings across methods (rows) and datasets (columns). **Rows:** HVG (raw highly variable genes), PCA, VAE, and Denoised (VAE + LDM). **Columns:** allCT_1.0k, group_5_subset_76, GSE114412_cluster, GSE114412_subcluster. Points colored by cell type. Dataset-specific hyperparameters (VAE latent dimension, PCA dimension, denoising steps) shown in column headers.

A UMAP of denoising effect for the best configuration of each of the datasets was also produced (figure 12). Visually inspection of this did not reveal a clear pattern of diffusion followed by denoising having an effect on cell type separation in UMAP embedding space.

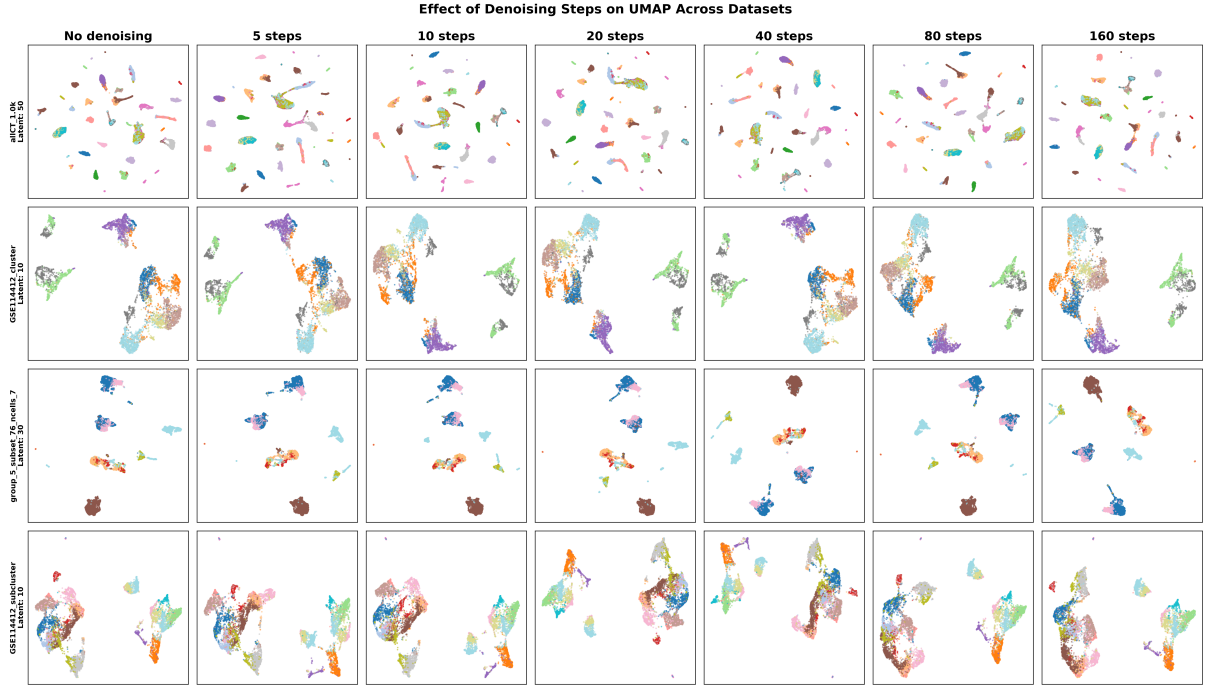


Figure 12: UMAP projections of VAE embeddings with progressive LDM denoising. Rows: datasets. Columns: denoising steps (0 = baseline VAE). Points colored by cell type.

4.10 Part 2 results summary

PCA generally outperformed VAE and LDM methods in preserving local cell type neighborhood structure (cLISI scores), while label transfer accuracy results were dataset-dependent. Increasing diffusion denoising steps did not consistently improve performance. The weak and variable correlation between cLISI and classification accuracy suggests these metrics capture complementary aspects of embedding quality.

5 Discussion

5.1 Part 1: Differential Expression in Homogeneous Cell Populations.

The first part of this thesis investigated how transcriptional homogeneity affected the performance of standard differential expression (DE) analysis for cell type characterization. The analysis was motivated by the observation that samples from cell differentiation protocols were often homogeneous, because of shared lineage in the samples. Therefore, to diagnose the effect, we systematically evaluated DE analysis performance across subsets of increasing transcriptional similarity. By constructing controlled subsets of cell types with defined similarity levels (see figure 2 and table 2), we were able to investigate the effect of homogeneity on downstream marker identification and cell type recovery.

Overall, the results demonstrate that the reliability of DE analysis based marker discovery degrades as samples becomes increasingly transcriptional similar. While DE analysis performed well for heterogeneous subsets containing clearly distinct cell types, its effectiveness decreased substantially in homogeneous settings, including subsets composed of closely related cell types and even subsets containing a single annotated cell type. These findings confirm that transcriptional homogeneity represents a fundamental challenge for DE-based cell type characterization, rather than a failure attributable to insufficient sample size or annotation noise alone.

Across the two metrics (Top-1 accuracy, Top-5 accuracy) DE performance was lowest in the most homogeneous subsets. Groups consisting of a single cell type (Groups 1 and 2) and groups composed of highly similar cell types (Groups 3–5) showed a high proportion of failed cell type matches in the gene

set enrichment analysis. As transcriptional diversity increased, DE performance improved steadily, reaching its highest accuracy in the highly diverse subsets.

These findings are particularly relevant in the context of cell differentiation, where transcriptional changes often occur gradually and lineage relationships impose substantial overlap in gene expression profiles. Progenitor cells, intermediate differentiation states, and closely related mature cell types frequently share transcriptional programs, differing primarily in subtle shifts in regulatory activity or expression magnitude rather than in the presence or absence of unique marker genes [36].

In such settings, the underlying signal DE tries to measure gets weakened. If transcriptional similarity increases and the differences are small and subtle, then the signal becomes weak and risks being drowned in the technical noise or differences in expressed housekeeping genes. As we see in the homogenous subsets, this aligns well with our results in which marker genes obtained from comparisons between closely related cells are not discriminative for the cell type.

These results emphasize that DE-based approaches are not well suited for discriminating between closely related cell states commonly encountered in differentiation studies. This does not imply that DE analysis is fundamentally flawed, but rather that its use is constrained by the biological structure of the data and the degree of transcriptional overlap between the populations under comparison.

However, the analysis has some limitations that should be acknowledged and addressed. First, transcriptional similarity was quantified using Pearson correlation between average expression profiles. While this measure captures global similarity and is commonly used in clustering and exploratory analyses [9], alternative similarity or distance metrics may emphasize different aspects of transcriptional variation. Nevertheless, the pattern observed between similarity and DE performance is unlikely to be different with the choice of metric. As we saw that the biologically homogenous motivated groups (Group 1 and 2), also were very homogeneous with the Pearson metric even though not computed with this (see figure 2).

Second, both the sampled subsets and the reference gene sets used for gene set enrichment analysis were derived from the DISCO atlas. This could mean that a potential dependency between the data used in the analysis to create the subsets and the data used for reference construction. This might inflate the absolute performance estimates. However, it does not explain the strong relative differences observed across similarity levels, which consistently follow the same trend.

Overall, these limitations does not seem to give rise to substantial worries that would invalidate the conclusion that transcriptional homogeneity poses a fundamental challenge for DE-based marker identification.

These findings, motivate the exploration of alternative strategies that could diminish noise and enhance signal. That is why we in the second part will focus on improving latent representaions of the data. In particular, meaningful clustering of cells is essential for enabling robust comparisons between cell populations, and such clustering depends on the quality of the embedding on which it is calculated. Accordingly, Part 2 of this thesis investigates whether alternative embedding methods can improve clustering quality and downstream interpretability in transcriptionally homogeneous settings, thereby addressing the limitations identified in Part 1.

5.2 Part 2: Evaluating Latent Representations

In Part 2, we did not find any evidence of VAE embeddings denoised by a DDPM provided a representation that better captured underlying biological structures compared to PCA in either the homogeneous datasets nor the heterogeneous. This became evident when evaluating cLISI and label transfer (LT) accuracy, where PCA the majority of cases outperformed the other methods. Starting out, we hypothesized that VAE and LDM would outperform PCA specifically in homogeneous datasets, namely the beta cell differentiation datasets GSE114412_cluster/subcluster and the DISCO subset (group_5_subset_76). However, this was not observed. Instead, all methods showed proportional changes in cLISI scores across datasets of varying homogeneity, with generally no preferential improvement of VAE or LDM in homogeneous settings.

While we observed a marginal improvement of VAE when in denoising in one special case—specifically for GSE11412_subcluster (Figure 6), where the average cLISI score across 10 runs increased from 0.9180 (VAE LD=40) to 0.9192 (LDM)—this difference is minimal (0.012% improvement) and not significant ($p > 0.05$), and does not constitute strong evidence for LDM superiority. Moreover, this apparent improvement may be an artifact of our model selection strategy. Rather than selecting the VAE latent dimension that achieved the highest cLISI score at denoising step 0 (Figure 5), we chose the latent dimension with the highest average cLISI across all denoising steps. This approach may have inadvertently excluded the optimal VAE configuration for the baseline (non-denoised) condition, potentially biasing the comparison in favor of the denoised embeddings of the LDM.

More specifically, we saw that especially when evaluating with cLISI score PCA is the better choice, but we did note that when evaluating with LT the VAE and HVG representations does outperform PCA for the group_5_subset_76 and the allCT_1.0k datasets. The reason why VAE might perform better on allCT_1.0k may be due to the high number of labels; with a large number of classes, it becomes harder for classifiers to find separating hyperplanes in linearly transformed spaces such as PCA. However, this does not explain why using HVG as a predictor also outperforms PCA.

The UMAP plots in Figure 12 demonstrated that denoising did not substantially improve the latent representation. Visual inspection revealed that the denoising process failed to relocate outlier cells into a more coherent intrinsic manifold beyond what was already captured by the VAE embeddings.

The minimal difference between VAE and denoised representations implies that the VAE latent space already provides a well-structured manifold with limited scope for further denoising improvements. This suggests the VAE encoder successfully learns to represent different expression profiles in structured clusters while suppressing noise during the initial compression, with diffusion-based denoising offering no refinements.

Dataset-specific patterns reveal differences in data quality and biological complexity. The group_5_subset_76_ncells_7 dataset’s consistent separation across methods indicates high-quality data with distinct cell populations, while allCT_1.0k’s persistent fragmentation suggests either genuine biological heterogeneity with many transitional states or unresolved technical variation. Notably, despite group_5_subset_76_ncells_7 dataset being selected as the most homogeneous dataset based on inter-cell-type Pearson correlation, the observed fragmentation suggests it may not be as homogeneous as intended—potentially due to high intra-cell-type heterogeneity or the presence of rare cell states not captured by the cell type-level correlation analysis. The GSE114412 datasets’ compact clusters indicate well-defined cell populations with minimal confounding factors.

Summing up, neither the VAE embeddings nor the LDM refinement led to improved distinguishability between the cells in latent space, when compared to the PCA. Thus, they most likely would not have increased subsequent grouping either. As the relative ranking of methods did not change across datasets, varying only in magnitude, there is no indication that the more complex VAE and LDM models perform relative better in homogeneous settings. This raises the question of whether the observed results stem from the data itself, the embedding methods, or the evaluation framework used to assess them.

5.3 Why VAE and LDM Did Not Outperform PCA

Both variational autoencoder and diffusion models have demonstrated strong performance in other domains like computer vision. In image analysis, iterative denoising through diffusion models have been shown to effectively reduce technical while enhancing resolution, for example in *Saharia et al (2023)* [16]. These models learn a low-dimensional manifold representation of the data and restore corrupted observations by iteratively denoising them toward this learned manifold.

However, as mentioned in the introduction, it is not obvious that the methods transfer to a biological domain. Gene expression data differ fundamentally from images in both structure, sparsity and noise. Nevertheless, given its strong performance in other domains it is still reasonable to hypothesize that it can refine latent representations in transcriptomic data. It should be noted that our model architecture for the LDM is fully adopted from scVAED [19], and in our thesis we did not fully explore the

hyperparameter space of latent dimensions and diffusion denoising steps. Beyond these hyperparameters, numerous other parameters within the VAE and LDM were left unchanged. While a systematic search through this parameter space might have improved model performance, the potential benefits remain unclear and would have required substantial additional computations.

Nevertheless, other studies have found similar observations as our results when benchmarking complex models for scRNA-seq analysis. They found that foundation models and transformer based architectures performed similar to, or worse, than simpler baseline methods in downstream tasks such as cell type classification [37, 38, 39]. Like our results suggested, these reviews also noted that limited or no gains were obtained from these complex models in scRNA compared to standard workflows [39].

However, in some cases, improved performance has been achieved by including biological structure or context, like in *Yuan, Xinyu et al. (2024)* [40], where a cell ontology information was added in the pretraining process of a foundation model. They similarly found that most complex models performed similar or worse to standard methods, while their method could outperform all the other models as well as baseline methods. This suggests that biological context (e.g. label ontology) is important especially when evaluating on labels derived from standard pipelines.

5.3.1 Evaluation metrics

To evaluate the quality of the learned representations of the models, we employed both the cLISI score and label transfer accuracy which both are commonly used metrics for quality assessment of latent representations in the literature. Also, the use of two complimentary metrics were used to mitigate any potential shortcomings with relying on a single evaluation criterion.

Across all datasets and methods, both metrics largely exhibited the same trends, that PCA performed better than the VAE and diffusion based methods. This agreement between metrics suggest that the observed performance is robust across evaluation, rather than an artifact of the evaluation metric. However, a critical limitation of both metrics is that they rely on good ground truth labels. The quality of the annotations therefore constrains what these metrics can reveal about the embeddings.

Because of the labels big influence on the evaluation of the embeddings it is useful to consider how the labels are generated in practice. The labels in the in-vitro dataset (GSE114412_subcluster, GSE1144112_cluster) were derived by initial clustering and then a subsequent marker analysis driven by expert interpretations. The clustering was constructed on a low dimensional latent representation from a PCA. Annotation by these steps are standard practice for the field [9] and used in many annotation pipelines.

Similarly, atlases like the Novo Nordisk atlas as well as the DISCO atlas set are created by integrating multiple datasets and harmonize cell type labels across studies. The atlases are often used as references for label transfer and regarded as containing high quality cell type representations, the original labels are often derived from some sort of clustering-based approach that rely on a dimensionality reduction like PCA [9]. As a result, many existing annotations may implicitly reflect separations, that are derived from a PCA.

If labels are mostly defined by separability in PCA space, then that may systematically favor PCA based representation in the evaluation. In a setting like this, nonlinear embeddings like VAE and LDM may not align as well with the annotation geometry, even if they capture biological meaningful structure. Not that the labels necessarily are incorrect, but rather that they are not independent enough for evaluating alternative embeddings.

This motivates a more broad question of whether cells are truly linearly separable in transcriptomic space. Because, if they are not, maybe the labels we have now are an approximation that is constrained by the embedding that were used to define them. If the second is the case then improvements in embedding may not be detectable using evaluation metrics based on labels alone.

Of course, there is also an alternative interpretation: rather than the labels and evaluation methods being a constraint, it may be that VAE and denoising-based methods do not generalize well to

biological settings. While this cannot be ruled out, we explore label-related confounding further below.

5.4 Limitations of current annotation practices

As mentioned, standard practice for cell typing in scRNA-seq data involves dimensionality reduction followed by clustering, with cell type annotations then assigned at the cluster level. A central assumption underlying this annotation pipeline is that cells that are close to each other in a low dimensional embedding (most commonly PCA) are transcriptionally similar in the original high-dimensional gene expression space. However, as highlighted in *Ovchinnikova et al (2020)* [41], dimensional reduction methods introduce distortions, and neighborhood relationships in low dimensional embeddings are only an approximation of similarities in the full higher dimensional feature space. They demonstrate in UMAP, tSNE and PCA, how some high dimensional neighborhoods are preserved while others are distorted and includes spurious proximity to transcriptional dissimilar cells or are separated from transcriptional similar cells.

These properties have direct implications for cell type annotation. Clustering is normally performed on a low dimensional representation, and cell types subsequently assigned at cluster level based on average differential expressed genes of that cluster. This annotation procedure thereby assumes that the cluster is internally homogeneous. However, as highlighted above this is not always the case. *Ovchinnikova et al (2020)* [41] even found through exploratory analysis that cells close to cluster boundaries in embedding space are often more similar in the full feature space to cells from another cluster, indicating wrong cluster boundaries. As a result, annotation at cluster level can lead to mislabeling at single-cell level, even when the cluster-leveled annotation is biological plausible.

Nevertheless, this annotation noise most likely just reflects that cell identity in scRNA-seq data is continuous and context-dependent, rather than discrete as the labels we impose on them [36]. This is particularly relevant when benchmarking lower dimensional embeddings. When the labels are derived from clustering in a specific latent space, then evaluation metrics that rely on these labels may favor those methods that reproduce the same embeddings space.

All in all, these observations highlight a general challenge in scRNA-seq analysis. Standard annotation practices are constrained by the geometry of the low dimensional embeddings and the assumption of within cluster homogeneity in cluster-based annotation. This makes it difficult to obtain cell type labels that are independent of the computational methods used to define them, which limits the interpretability of downstream benchmarking.

5.5 How to get better ground truth and evaluation

Improving cell type annotations would require experimental strategies that provide stronger ground truth identities independent of representational learning tools like PCA. However, single-cell data presents fundamental challenges for annotation. Unlike domains such as image classification, where non-experts can readily distinguish a dog from a cat, individual gene expression vectors are not directly interpretable by human annotators. Even in computer vision, where visual inspection is straightforward, ambiguous cases persist—the "Chihuahua or muffin" problem (figure 13) illustrates how superficial similarities can confuse classification even for humans [42]. If such confusion occurs in a domain with relatively clear perceptual features, the challenge is far greater for transcriptomic data, where distinguishing, for example, pancreatic alpha cells from pancreatic beta cells requires specialized knowledge and may still be ambiguous even for experts.

Moreover, in scRNA-seq analysis, individual cell labels are typically not directly human-annotated. Instead, they represent cluster-level characteristics assigned to clusters of cells deemed similar in embedding space through computational clustering. This means annotations reflect the geometry of the embedding method used, rather than independent biological verification at single-cell resolution. Consequently, scRNA-seq analysis inherently involves uncertainty in ground truth labels.

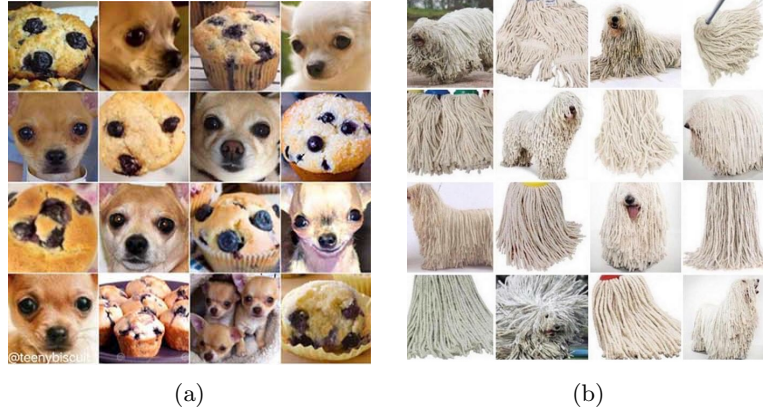


Figure 13: (a) and (b) are examples of how label ambiguity can occur even in domains where ground truth labels are abundant and seemingly certain. Source: [43, 44]

However, methods like micro-dissection-based sampling where tissue or cell identity is known a priori [45], or utilizing different complimentary modalities such as spatial transcriptomic or multi-omics [46], could help make cell identity labels less dependent on transcriptomic data and the accompanying computational methods alone. Ideally the labels should not be solely computer based.

Alternatively, an evaluation method that does not rely on labels could be explored. For instance, embeddings could be assessed by biological coherence between marker genes identified through differential expression analysis. Biological coherence could be interpreted through gene set enrichment analysis (GSEA), where a big overlap in marker genes and reference gene sets would indicate high biological coherence. However, such approach would introduce its own challenges as it lack a clear ground truth.

Taken together, these observations suggest that cell typing and creating good latent representations in scRNA-seq data constitute an inherently difficult problem. The data are noisy, high-dimensional, and biologically complex, and ground truth labels are often derived through computational procedures that may bias evaluation. In this context, the lack of improvement from more complex embedding methods does not necessarily indicate methodological failure. Rather, it reflects structural limitations in current annotation practices and benchmarking frameworks, which constrain the ability to detect genuine improvements in representation quality.

At the same time, recent work has demonstrated that incorporating biological context into modeling frameworks can improve performance. For example, Yuan et al. [40] showed that integrating cell ontology information, into complex model makes it successfully outperform baseline methods. Similarly, several diffusion-based models for scRNA-seq incorporate conditioning on biological metadata such as cell type, tissue of origin, or developmental stage [17, 18]. These findings suggest that model expressiveness alone may be insufficient without explicit incorporation of biological context.

5.6 Future research

Future work could focus on systematically assessing the quality and consistency of cell type annotations in scRNA-seq datasets, particularly at the single-cell level. Improved understanding of annotation uncertainty would provide valuable context for benchmarking representation learning methods.

It would also be interesting to see if incorporating biological meta data in the modeling framework could improve representational performance and downstream tasks. As mentioned earlier, recent studies suggest that including contextual information can help complex models capture biological meaningful structure.

In addition, alternative evaluation paradigms could be explored that avoid reliance on discrete cell type

labels altogether. One promising direction would be to define reference neighborhoods or clusters in an intermediate-dimensional space (e.g., 100–1000 dimensions), which may mitigate the curse of dimensionality while retaining sufficient biological structure. Low-dimensional embeddings could then be evaluated based on how well they preserve these intermediate-scale neighborhoods. Such an approach could reduce circularity of evaluation, provide a continuous measure of information preservation across dimensionality scales, and reveal whether certain embedding methods systematically distort specific aspects of biological structure during dimensionality reduction.

6 Conclusion

This thesis investigated how transcriptional heterogeneity influence cell type characterization in single-cell RNA sequencing data, and whether alternative latent representations can help mitigate the limitations observed in standard classification workflows. Across two parts, the thesis first examined the influence of sample heterogeneity on marker discovery through differential expression analysis and secondly potential benefits and limitations of nonlinear embedding and refinement strategies.

In Part 1, we demonstrated that standard differential expression (DE) analysis performance decreases systematically as samples becomes increasingly transcriptional similar. Reliant cell type classification through DE-based marker discovery performed robustly in heterogeneous where samples contains clearly distinct cell types. For homogeneous settings, where samples contains transcriptional similar cell types, standard DE analysis did not identify relevant marker genes for subsequent cell type classification. These findings highlight and confirms that marker-based cell type annotation in homogeneous samples posses a fundamental challenge, why careful considerations need to be taken when working with homogeneous samples.

In Part 2, we explored whether alternative latent representations could improve downstream cell grouping and ultimately classification in homogeneous settings. We evaluated three increasingly complex embedding strategies, ranging from principal component analysis (PCA), Variational Autoencoder (VAE) and finally a Latent Diffusion Model (LDM). All embedding strategies were tested against a baseline of a selection of normalized highly variable genes (HVG). The analysis, showed that even though all methods represented the data in a meaningful way, increased model complexity did not increase performance. The linear latent representation PCA performed slightly but consistently better than VAE and LDM in both homogeneous and heterogeneous curated data sets. These findings suggest that the nonlinear and generative methods assessed does not recover additional biological relevant structure, under the evaluated conditions. The results align with evaluation studies of other complex models like foundation models. All in all, the results indicates that the limitations in homogeneous settings observed in Part 1, is not trivially solved by more expressive models.

Together, the results highlight that challenges in homogeneous single-cell data arise not only from modeling limitations but also from annotation and evaluation practices that rely on labels derived from standard clustering pipelines. When ground truth labels are uncertain or method-dependent, improvements in latent representations may remain undetectable. More broadly, this thesis underscores the intrinsic difficulty of cell type characterization in transcriptionally similar populations and emphasizes the need for improved annotation frameworks and evaluation strategies to enable meaningful methodological progress.

7 Code availability

Code can be found at <https://github.com/AntonWangDTU/Special>

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A DISCO data

Table 3: Number of unique cell types per DISCO tissue dataset.

Tissue	# Cell Types
adipose	36
adrenal gland	11
bladder	36
blood	24
bone marrow	54
brain	49
breast	32
embryo	81
esophagus	31
eye	24
fallopian tube	23
heart	27
intestine	57
kidney	50
liver	26
lung	58
ovary	39
pancreas	30
prostate	33
skin	29
spleen	20
stomach	30

B Novo Data information

Table 4: SING Dataset Overview

Dataset	Species	Tissue	Data Source	Cells Profiled	Data Description Link
4 (UMI)	Human	Liver	Public	58,000	Liver cirrhosis is a major cause of death worldwide and is characterized by extensive fibrosis. There are currently no effective antifibrotic therapies available. To obtain a better understanding of the cellular and molecular mechanisms involved in disease pathogenesis and enable the discovery of therapeutic targets, here we profile the transcriptomes of more than 100,000 single human cells, yielding molecular definitions for non-parenchymal cell types that are found in healthy and cirrhotic human liver. https://www.ncbi.nlm.nih.gov/pubmed/31597160

Continued on next page

Table 4 – Continued from previous page

Dataset	Species	Tissue	Data Source	Cells Profiled	Data Description Link
9 (UMI)	Human	Heart	Public	6,371	They studied the cell composition of normal human heart by single-cell sequencing. Distinct subgroups of cardiac muscle, fibroblast cell and endothelial cell were detected. https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE109816
25	Human	Liver	Public	15,387	This data examines the heterogeneity and dynamic crosstalk of human liver cells, 18,000 freshly isolated human liver cells from 6 donors were profiled (E-MTAB-10553).
26	Human	Brain; Substantia nigra	Public	6,032	Single-nuclei RNAseq data of the substantia nigra brain region from 5 human (4x male, 1x female) donors post-mortem. Age 55-70. Astrocytes, microglia and in part neurons were reannotated relative to original paper, as sample-specific clusters were detected. Agarwal, 2020, Nature Commun.
27	Human	Brain; Cortex	Public	10,758	Single-nuclei RNAseq data of the brain cortex from 5 human (4x male, 1x female) donors post-mortem. Age 55-70. This dataset contains samples that are pair-sampled with the substantia nigra data, also from Agarwal et al. Cell type clustering is identical to the original paper (Figure 1B). Agarwal, 2020, Nature Commun.
34	Human	Adipose; visceral and subcutaneous	Public	166,149	A single-cell atlas of human white adipose tissue. Emont et al. 2022, Nature
44	Human	Pancreatic Islet	Public	222,077	Single-cell RNA-seq of human pancreatic islets from healthy, normoglycemic pancreas auto-antibody positive (AAB), T1D and T2D donors. The dataset is available through the Human Pancreas Analysis Program. https://hpap.pmacs.upenn.edu/
45	Human, Pig, Mouse	Pancreatic Islet	Public	36,605	A transcriptional cross-species map of human, mouse and porcine pancreatic beta cells. https://pubmed.ncbi.nlm.nih.gov/36113773/
46	Human, Pig, Mouse	Pancreatic Islet	Public	14,178	A transcriptional cross-species map of human, mouse and porcine pancreatic alpha cells. https://pubmed.ncbi.nlm.nih.gov/36113773/

Continued on next page

Table 4 – *Continued from previous page*

Dataset	Species	Tissue	Data Source	Cells Profiled	Data Description Link
54	Human	Heart	Public	191,795	High-resolution map of human cardiac remodelling after myocardial infarction using single-nucleus RNA-seq in myocardium from patients with myocardial infarction (16) and controls (4). Kuppe et al. 2022, Nature. https://pubmed.ncbi.nlm.nih.gov/35948637/
64	Human	Heart	Public	55,122	Regulatory changes in hypertrophic cardiomyopathy unraveled by single-nucleus RNA-seq. Liu et al. 2023, Cell Discov. https://pubmed.ncbi.nlm.nih.gov/36646705/
72	Human	Heart	Public	451,513	Cells of the adult human heart. Single-nucleus RNA-seq on heart samples from 14 donors, 7 female and 7 male within the age range of 40-75. Litvinukova et al. 2020, Nature. https://pubmed.ncbi.nlm.nih.gov/32971526

C VAE and Diffusion Architecture and Parameters

Table 5: Diffusion model and denoising neural network architecture

Component	Specification
Time Embedding (Learned)	
Embedding type	Learned embedding layers
Number of embeddings	$T = 2000$ (one per timestep)
Embedding dimension	512 (hidden dimension)
Integration	Additive (summed with hidden states)
Noise Prediction Network	
Input dimension	Latent dimension
Hidden dimension	512
Number of layers	4
Layer structure	Linear $\rightarrow$ + Time Emb $\rightarrow$ ReLU (repeated)
Output dimension	Latent dimension
Activation	ReLU
Training Stabilization	
Exponential Moving Average	Applied to model parameters
EMA decay rate (μ)	0.9999
Diffusion Process	
Total timesteps (T)	2000
Noise schedule	Sigmoid
β range	$[10^{-5}, 10^{-2}]$

Table 6: VAE model architecture and hyperparameters

Component	Specification
Encoder	
Input	Log-normalized HVG expression
Hidden layers	Linear(input_dim, 100) $\rightarrow$ ReLU
Latent projections	Linear(100, latent_dim) for μ and $\log \sigma^2$
Latent Space	
Distribution	$z \sim \mathcal{N}(\mu(x), \sigma^2(x))$
Sampling	$z = \mu + \epsilon \odot \exp(0.5 \cdot \log \sigma^2)$, $\epsilon \sim \mathcal{N}(0, I)$
Dimensions tested	10, 20, 30, 40, 50
Decoder	
Hidden layers	Linear(latent_dim, 100) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.2)
Output layer	Linear(100, input_dim)
Output space	Reconstructed log-normalized expression
Loss Function	
Reconstruction	$\mathcal{L}_{\text{recon}} = \text{MSE}(x, \hat{x})$
KL divergence	$\mathcal{L}_{\text{KL}} = -0.5 \sum_{i=1}^d (1 + \log \sigma_i^2 - \mu_i^2 - \sigma_i^2)$
Total loss	$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \mathcal{L}_{\text{KL}}$
β weight	1.0

D Label Transfer

D.1 Label Transfer Evaluation with Logistic Regression

To assess the biological utility of different data representations, we performed cell type label transfer using multinomial logistic regression. This supervised learning task evaluates how well each representation preserves cell type identity information that can be leveraged for classification.

D.2 Data Preparation and Split

The dataset was split into three independent subsets: training (used to fit the classifier), validation (used for hyperparameter selection), and test (used for final performance evaluation). To prevent data leakage during preprocessing, highly variable gene (HVG) selection was performed exclusively on the training set. Specifically:

1. Each split was independently normalized to a median total count of 10^4 per cell
2. Log-transformation was applied: $X' = \log(X + 1)$
3. The top 2,000 highly variable genes were identified from the training set only
4. All three splits were then subset to these same 2,000 genes

This ensures that the validation and test sets remain truly independent, with no information from these sets influencing the feature selection process.

D.3 Representations Evaluated

We evaluated four categories of data representations:

- **Raw HVG:** Preprocessed gene expression data (2,000 HVGs)
- **PCA:** Principal component representations with $d \in \{10, 20, 30, 40, 50\}$ dimensions
- **VAE:** Variational autoencoder latent representations with $d \in \{10, 20, 30, 40, 50\}$ dimensions
- **Denoised:** VAE-denoised gene expression with latent dimension $d \in \{10, 20, 30, 40, 50\}$ and denoising steps $s \in \{50, 100, 150, 200, 250\}$, resulting in 25 configurations per latent dimension

D.4 Classification Model

For each representation, we trained a multinomial logistic regression classifier using the following configuration:

$$P(y = k|\mathbf{x}) = \frac{\exp(\mathbf{w}_k^T \mathbf{x} + b_k)}{\sum_{j=1}^K \exp(\mathbf{w}_j^T \mathbf{x} + b_j)} \quad (12)$$

where $\mathbf{x}$ is the input representation, K is the number of cell type classes, and $\mathbf{w}_k, b_k$ are the learned weights and bias for class k .

The model was trained using the Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) optimizer with a maximum of 1,000 iterations. L-BFGS is particularly well-suited for this problem as it efficiently handles the high-dimensional parameter space while maintaining convergence stability for multinomial classification [?].

D.5 Performance Metrics

We evaluated classification performance using multiple complementary metrics on both validation and test sets:

- **Accuracy:** Overall proportion of correctly classified cells

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_i = \hat{y}_i) \quad (13)$$

- **Weighted F1 Score:** Harmonic mean of precision and recall, weighted by class support to account for class imbalance

$$F1_{\text{weighted}} = \sum_{k=1}^K \frac{n_k}{N} \cdot \frac{2 \cdot \text{precision}_k \cdot \text{recall}_k}{\text{precision}_k + \text{recall}_k} \quad (14)$$

- **Macro F1 Score:** Unweighted average F1 score across all classes, giving equal importance to rare and common cell types

$$F1_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{2 \cdot \text{precision}_k \cdot \text{recall}_k}{\text{precision}_k + \text{recall}_k} \quad (15)$$

- **Balanced Accuracy:** Average of recall scores per class, compensating for imbalanced class distributions

$$\text{Balanced Accuracy} = \frac{1}{K} \sum_{k=1}^K \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k} \quad (16)$$

where N is the total number of samples, n_k is the number of samples in class k , and TP_k and FN_k are true positives and false negatives for class k , respectively.

D.6 Integration with Batch Mixing Metrics

To identify optimal representations that balance both biological signal preservation (measured by label transfer performance) and batch effect removal (measured by cLISI scores), we computed the Pearson correlation coefficient between test set performance metrics and cLISI scores across all denoised configurations. This analysis reveals whether configurations achieving superior batch mixing also maintain discriminative cell type information, or if there exists a trade-off between these two objectives. Statistical significance of correlations was assessed using two-tailed t-tests, with significance levels indicated as: *** for $p < 0.001$, ** for $p < 0.01$, and * for $p < 0.05$.

E FindAllMarkers() function

FindAllMarkers() from the Seurat package [4] were used in the DE analysis in part 1 of the thesis. The function were used with the following parameters:

- *only.pos = True* - We only kept unregulated marker genes
- *min.pct = 0.1* - This parameter indicates the minimum fraction of cells that needs to have expressed the gene in either of the populations, before testing the gene for differential expression.
- *logfc.threshold = 0.25* - Only test genes that on average show at least X-fold (log-scale) between the groups of cells.

Elaborations on parameters and parameterization can be found here

F Statistical Test of Embedding Methods

G Subset group examples

Comparison	p_{adj}	Mean diff.	Signif.	Higher
allCT_1.0k				
LDM > HVG	0.9219	0.000108	No	LDM
PCA > LDM	0.8633	0.000595	No	PCA
PCA > HVG	0.8633	0.000703	No	PCA
PCA > VAE	0.9219	0.000361	No	PCA
VAE > LDM	0.8633	0.000234	No	VAE
VAE > HVG	0.9219	0.000342	No	VAE
GSE114412_cluster				
LDM > HVG	0.0039	0.037787	Yes	LDM
LDM > VAE	0.8457	0.000985	No	LDM
PCA > LDM	0.0293	0.016636	Yes	PCA
PCA > HVG	0.0039	0.054423	Yes	PCA
PCA > VAE	0.0328	0.017621	Yes	PCA
VAE > HVG	0.0039	0.036803	Yes	VAE
group_5_subset_76_ncells_7				
LDM > VAE	0.8457	0.000615	No	LDM
PCA > LDM	0.0039	0.021344	Yes	PCA
PCA > HVG	0.0234	0.0107	Yes	PCA
PCA > VAE	0.0039	0.021959	Yes	PCA
HVG > LDM	0.0146	0.010644	Yes	HVG
HVG > VAE	0.0039	0.011259	Yes	HVG
GSE114412_subcluster				
LDM > HVG	0.0039	0.015542	Yes	LDM
PCA > LDM	0.0059	0.010039	Yes	PCA
PCA > HVG	0.0039	0.025581	Yes	PCA
PCA > VAE	0.0164	0.00851	Yes	PCA
VAE > LDM	0.4316	0.001529	No	VAE
VAE > HVG	0.0039	0.017071	Yes	VAE

Table 7: Pairwise Wilcoxon signed-rank tests (FDR-adjusted) for cLISI within each dataset. Mean diff. follows the comparison direction (e.g., “VAE > HVG” reports $\bar{\Delta} = \text{VAE} - \text{HVG}$).

Comparison	p_{adj}	Mean diff.	Signif.	Higher
allCT_1.0k				
LDM > PCA	0.0098	0.0101	Yes	LDM
HVG > LDM	0.0023	0.0348	Yes	HVG
HVG > PCA	0.0023	0.0449	Yes	HVG
HVG > VAE	0.0023	0.0312	Yes	HVG
VAE > LDM	0.0023	0.0036	Yes	VAE
VAE > PCA	0.0023	0.0137	Yes	VAE
GSE114412_cluster				
LDM > VAE	0.8203	0.0005	No	LDM
HVG > LDM	0.0029	0.0308	Yes	HVG
HVG > VAE	0.0029	0.0313	Yes	HVG
PCA > LDM	0.0029	0.0416	Yes	PCA
PCA > HVG	0.0164	0.0108	Yes	PCA
PCA > VAE	0.0029	0.0421	Yes	PCA
group_5_subset_76_ncells_7				
LDM > PCA	0.0059	0.0155	Yes	LDM
HVG > LDM	0.4312	0.0026	No	HVG
HVG > PCA	0.0078	0.0181	Yes	HVG
HVG > VAE	0.5703	0.0009	No	HVG
VAE > LDM	0.041	0.0017	Yes	VAE
VAE > PCA	0.0059	0.0172	Yes	VAE
GSE114412_subcluster				
HVG > LDM	0.0029	0.0439	Yes	HVG
HVG > VAE	0.0029	0.041	Yes	HVG
PCA > LDM	0.0029	0.0489	Yes	PCA
PCA > HVG	0.3125	0.005	No	PCA
PCA > VAE	0.0029	0.046	Yes	PCA
VAE > LDM	0.0258	0.0029	Yes	VAE

Table 8: Pairwise Wilcoxon signed-rank tests (FDR-adjusted) for label transfer accuracy within each dataset. Mean diff. follows the comparison direction (e.g., “VAE > PCA” reports $\bar{\Delta} = \text{VAE} - \text{PCA}$).

Group	Subsets (tissues / cell types)	Mean cor	n_cells
1	heart_ Dendritic cell	NA	1
2	blood_ Naive B cell ; bone marrow_ Naive B cell ; intestine_ Naive B cell ; liver_ Naive B cell ; lung_ Naive B cell	0.9345	5
2	adipose_ Lymphatic EC ; bladder_ Lymphatic EC ; breast_ Lymphatic EC ; mouth_ Lymphatic EC	0.8561	4
3	lung_ Arterial EC ; lung_ Capillary EC	0.9489	2
4	lung_ AT2 ; lung_ Cycling AT2 ; lung_ AT1	0.9320	3
5	pancreas_ Alpha cell ; pancreas_ Beta cell	0.9343	2
6	blood_ Memory CD8 T cell ; tonsil_ Naive CD8 T cell	0.8124	2
8	breast_ pDC ; testis_ Elongated spermatid	0.2159	2
9	liver_ Hepatocyte ; brain_ Oligodendrocyte ; lung_ Alveolar macrophage	0.4682	3

Table 9: Example selected subsets illustrating group size (n_cells) and within-subset similarity (mean_cor). Nodes list the tissue_celltype labels included in each subset; the cell type portion is bolded for readability.